TERMINATION OF REAL FLIPS IN DIMENSION FOUR

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Abstract. OpenAI recently claimed a proof of the termination of flips for fourfold projective log canonical pairs with rational coefficients, but the relative case and the real coefficients case were not proved. In this paper we prove the termination of (relative) flips for all fourfold log canonical pairs, hence the termination of (relative) MMP for all fourfold log canonical pairs.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

The minimal model program for lc pairs consists of three ingredients: the cone and contraction theorems, the existence of flips, and the termination of the minimal model program. The cone and contraction theorems for lc pairs were proved in [Fuj11, Theorem 1.1], and lc flips exist by [Bir12, Corollary 1.2] and [HX13, Corollary 1.8]. Thus we can run the minimal model program for lc pairs, and the remaining problem is its termination.

Conjecture 1.1. *Let $(X, B)/U$ be an lc pair. Then any $(K_X + B)$-MMP/U terminates.*

We may lift any lc MMP to a $\mathbb{Q}$-factorial dlt MMP (cf. [Bir12, Remark 2.9]). Since the Picard number decreases under each divisorial contraction of a $\mathbb{Q}$-factorial MMP, Conjecture 1.1 is equivalent to the termination of flips.

Conjecture 1.2. *Let $(X, B)/U$ be an lc pair. Then any sequence of (K_X+B)-flips/U terminates.*

By [OAI26a], $(X, B)/U$ has a good minimal model or a Mori fiber space, so any $(K_X + B)$-MMP/U with scaling of an ample divisor terminates (cf. [BCHM10], [Bir12, Theorem 1.9]). In particular, *some* $(K_X + B)$-MMP/U terminates. This does not imply Conjecture 1.2, which predicts that *any* sequence of $(K_X + B)$-flips/U terminates.

Conjecture 1.2 was known when $\dim X \leq 3$ [Sho85, Kaw92, Sho96], and it was open when $\dim X \geq 4$. When $\dim X = 4$, most cases were known:

(1) When X is $\mathbb{Q}$-factorial terminal and $B = 0$ [KMM87].
(2) When (X, B) is projective canonical, B is a $\mathbb{Q}$-divisor, and either $\lfloor B \rfloor = 0$ or X is $\mathbb{Q}$-factorial [Fuj04, Fuj05]. The proofs of [Fuj04, Fuj05] work verbatim for $\mathbb{R}$-pairs.

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(3) When (X, B) is klt and $K_X + B$ is anti-effective/U [AHK07].
(4) When $K_X + B$ is effective [Bir07, HMX14].
(5) When the lc dimension of $(X, B)/U$ is ≥ 2 [Bir10b].
(6) When $K_X + B$ is pseudo-effective [CT23, Mor25].
(7) When (X, B) is klt and B is big/U [HLZ25, Corollary 3.2(3)], [Has26].
(8) Assuming the ACC and the lower-semicontinuity conjectures for minimal log discrepancies in dimension 4 [Sho04].
(9) Assuming the ACC for global minimal log discrepancies for enc singularities in dimension 4 [HL25].

Recently, [OAI26b] claimed a proof of Conjecture 1.2 when $\dim X = 4$, X is projective, and B is a $\mathbb{Q}$-divisor. For $\mathbb{R}$-divisors B, Conjecture 1.2 was still open when $\dim X = 4$, even when X is projective. The difference between $\mathbb{R}$-divisors and $\mathbb{Q}$-divisors looks technical, but it cannot be ignored: when $\dim X = 3$, the case when B is a $\mathbb{Q}$-divisor was proved in [Kaw92], while the case when B is an $\mathbb{R}$-divisor was proved much later in [Sho96]. In this paper, we prove Conjecture 1.2, hence Conjecture 1.1, in full generality when $\dim X = 4$:

Theorem 1.3. *Let $(X, B)/U$ be an lc pair of dimension 4. Then any sequence of steps of a $(K_X + B)$-MMP/U terminates. In particular, any sequence of $(K_X + B)$-flips/U terminates.*

The proof of Theorem 1.3 does not rely on [OAI26b, Theorem 1.2], but it follows the method of the proof of [OAI26b, Theorem 1.2], and we overcome the difficulties caused by the non-projectivity of the ambient variety and by the irrational coefficients.

Remark 1.4. In the main body of this paper, whenever we need a result of [OAI26b] or a variation of it, we state the result, give a proof which does not depend on [OAI26b], and cite the corresponding result of [OAI26b]. This is because the wording of [OAI26b] is unreadable, and the author feels that the birational geometry community should have a self-contained and well-written proof.

Sketch of the proof. Our strategy follows [AHK07]: we construct a difficulty function, show that it is monotone and bounded, and show that it decreases by at least a fixed positive number under $(1, 2)$-flips and $(2, 2)$-flips; the $(2, 1)$-flips are handled by the strict monotonicity of ρ_c. The difficulty function in [AHK07] is not fixed: it depends on a *weight* function, which is an $\mathbb{R}$-valued function satisfying [AHK07, Conditions 2.1].

Changing the weight function gives some flexibility, and many experts have tried to use it (cf. [Ale04]), but this has not led to the termination of all fourfold flips. It seems that any such approach needs at least the ACC for mlds of enc pairs (cf. [HL25]), which is still unknown.

This paper, as well as [OAI26b], follows this strategy, with one significant difference: our weight function is integer-valued (Definition-Lemma 5.2), hence so is the difficulty function (Definition-Proposition 5.5). This removes the key issue of finding a gap in the change of the difficulty function. Although the weight function is defined differently, the difficulty function is built from it in a very similar way: compare Definition-Proposition 5.5 with [AHK07, Definition 2.3].

In the author's view, the new difficulty function works, while the one in [AHK07] does not, because the latter loses some information. In [AHK07, Definition 2.2], the weight function is

$$w^-_\alpha(x) := 1 - x \text{ for } x \leq \alpha, \quad w^-_\alpha(x) = 0 \text{ for } x > \alpha,$$

or a function w^+_α defined in a similar way, where α is taken at the end of the proof to be the minimum of certain coefficients. What is lost is the set of coefficients of B: the fact that these coefficients belong to a finite set is not used at all. Our weights w_Γ depend on the set Γ generated by the coefficients of B, so the difficulty function in this paper uses this information.

This comes at a price. The difficulty function in [AHK07] is decreasing and is easily seen to be non-negative [AHK07, Lemma 2.6]. Our difficulty function is still decreasing (Proposition 5.8), but its boundedness is much harder to prove, and the new input is Hodge theory. We sketch the

idea here and refer the reader to Section 6 for details. The key is a close look at the difficulty D_{n_i} of the terminalization $X'_{n_i} \to X_i$ (Definition-Proposition 5.5).

The difficulty D_{n_i} is the sum of three terms. The second term is a sum over the irreducible surfaces on X'_{n_i}, and it is the only term which can be negative. Geometrically, it is negative only along the surfaces where the exceptional divisors/X_i are glued, and it is at least

$$-\sum_T (r_T - 1),$$

where T runs over the irreducible surfaces on the exceptional divisors/X_i and r_T is the number of branches of the exceptional divisors along T (Lemma 5.7). This negative term is compensated by the first and the third terms:

- The first term is at least the number of prime divisors E such that $\operatorname{center}_{X'_{n_i}} E$ is a curve on the exceptional divisors/X_i and $a(E, X'_{n_i}, B'_{n_i}) < 2$. Geometrically, it measures the singular curves of X'_{n_i} on the exceptional divisors/X_i.
- The third term is at least the number of independent classes of surfaces on the normalizations of the exceptional divisors/X_i.

This works if the exceptional divisors/X_i are contracted to closed points of X_i. Geometrically, each gluing along a surface, which contributes to the second term, corresponds either to an independent class of surfaces on the exceptional divisors/X_i, which contributes to the third term, or to a singular curve of X'_{n_i} on these divisors, which contributes to the first term.

A rigorous proof needs Hodge theory. A gluing along a surface which does not correspond to an independent class of surfaces produces a class of weight 4 in H^5 of the exceptional locus. The decomposition theorem shows that no such class exists when X'_{n_i} is smooth, and the intersection complex shows that a singular curve produces at most as many such classes as the prime divisors of log discrepancy 2 over it (Proposition 6.7). See Section 6 for details.

Finally, we compare our proof with the proof of [OAI26b], which treats projective pairs with rational coefficients, while we treat the relative case for pairs with arbitrary real coefficients. This change is minor but not trivial (compare [Kaw92] with [Sho96] in dimension 3). The main differences are the following.

(1) *The difficulty function.* The difficulty function in [OAI26b] uses the positive integer N such that Nb is an integer for any coefficient b of B [OAI26b, (4.1)]. For real coefficients, such an N does not exist. Instead, we take the set Γ of $\mathbb{N}$-linear combinations of the coefficients of B, and define the weights w_Γ, hence the difficulty function, using Γ. See Definition-Lemma 5.2 for details.

(2) *The boundedness of the difficulty function.* When X is projective, the difficulty function is non-negative [OAI26b, Proposition 7.2]. For relative MMPs, the proof is much more complicated.

We found two approaches, and we present one of them in this paper. In the other approach, we add weights coming from incomplete curves. Then we could not show that the difficulty function is non-negative, but we could still show that it is bounded from below by a (possibly negative) integer, which is enough for the termination of flips. We do not present this approach, as it involves more complicated Hodge theory.

The approach of this paper is simpler. The key is to reduce the question to the termination of *very sloppy flips* (Definition 4.2), for which the centers of the "bad exceptional divisors" are closed points. This resolves the issue of incomplete curves and simplifies the proof. The reduction needs the ACC for mlds in dimension 3, which is not known in full generality: a recent AI-generated result of the author [Liu26] passed the internal verifier but has yet to be verified by a human, so it may not be considered a human-verified proof. However, we only need the ACC for mlds of enc singularities [HL25, Theorem 1.6, Corollary 1.7]; see Lemma 4.5. Note that [OAI26b] only uses the ACC for mlds in dimension 2. We also remark that we cannot first prove the projective

case and then use a compactification, as a sequence of flips may no longer be a sequence of flips after compactification.

Remark 1.5. A similar construction of the difficulty function may lead to a proof of the termination of flips for all NQC generalized pairs in dimension 4. We leave this question to the reader (or perhaps to the reader's AI system).

A more interesting and more important question is the termination of flips, even with scaling, for generalized pairs which may not be NQC. For lc generalized pairs that are not necessarily NQC, the cone and contraction theorems hold by [CHLX23, Theorem A(1)] and flips exist by [HL26, Theorem 1.1], so such an MMP can be run [HL26, Theorem 1.2]. This question is closely related to the transcendental minimal model program for Kähler generalized pairs [HX26]. Up to now, we do not even know the existence of minimal models for klt non-NQC generalized pairs in dimension 3.

It seems that the arguments in this paper work verbatim for non-projective MMPs, that is, MMPs for quasi-projective pairs (X, B) without a fixed base U, but we do not consider them in this paper. Finally, it is interesting to consider the effective termination [HLZ25] for all fourfold pairs. This would make the paper much more complicated and its structure much less clear, so we also leave it to the reader.

Structure of the paper. In Section 2 we recall preliminaries, in particular the homology groups used in this paper and some basic properties of flips. In Section 3 we reduce the question to the termination of $\mathbb{Q}$-factorial klt flips and show that there does not exist an infinite sequence of $(2, 1)$-flips. In Section 4 we further reduce the question to very sloppy flips and construct a terminalization for each pair in the sequence. In Section 5 we construct the difficulty function from these terminalizations and show that it is well-defined, integer-valued, and monotonically decreasing. In Section 6 we show that the difficulty function is bounded from below. In Section 7 we prove the main theorem.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Basic minimal model program.

Definition 2.1. A *contraction* is a projective surjective morphism $f : X \to Y$ between normal quasi-projective varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Y$. A *birational morphism* in this paper is always assumed to be a contraction that is birational.

Definition 2.2. Let $f : X \to Y$ be a birational morphism. We denote by $\mathrm{Ex}(f)$ the exceptional locus of f and denote by $\mathrm{Exc}(f)$ the reduced divisor supported on $\mathrm{Ex}(f)$.

Definition 2.3. A *pair* $(X, B)/U$ consists of a projective morphism $\pi : X \to U$ from a normal quasi-projective variety X to a variety U, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that $(X, B)/U$ is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). The *discrepancy* of E with respect to (X, B) is $a(E, X, B) - 1$.

For any (sub-)pair $(X, B)/U$, we say that $(X, B)/U$ is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of $(X, B)/U$ is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of $(X, B)/U$ is the center of an lc place of $(X, B)/U$ on X. For any (not necessarily closed) point $x \in X$, we denote by

$$\mathrm{mld}(X \ni x, B) := \inf\{a(E, X, B) \mid E \text{ is a prime divisor over } X, \mathrm{center}_X E = \bar{x}\}$$

the *minimal log discrepancy* (*mld* for short) of $(X \ni x, B)$.

If U is not important, then we may omit U in the definitions above.

We shall use the following definition of a step of MMP in this paper. Note that our definition is slightly different from the classical definition of a step of MMP.

Definition 2.4. Let $(X, B)/U$ and $(X', B')/U$ be lc pairs. A *step of MMP*$/U$ $\phi : (X, B) \dashrightarrow (X', B')$ is of the form $X \xrightarrow{f} Z \xleftarrow{g} X'$ satisfying the following.

(1) f and g are birational morphisms$/U$.
(2) f is a contraction of a $(K_X + B)$-negative extremal ray.
(3) $\phi : X \dashrightarrow X'$ is the ample model$/Z$ of $(K_X + B)$.
(4) $B' = \phi_* B$.

If, in addition,

(5) f is small,

then we say that ϕ is a $(K_X + B)$-*flip*$/U$. A *sequence of steps* (resp. *sequence of flips*) of a $(K_X + B)$-MMP$/U$ is of the form

$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$

where each $\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1})$ is a step of a $(K_{X_i} + B_i)$-MMP$/U$ (resp. $(K_{X_i} + B_i)$-flip$/U$). If U is a closed point, then we may omit U in the definitions above.

Remark 2.5. Note that in the definition above, a Mori fiber space is not formally a step of MMP. On the other hand, we do not need to consider Mori fiber spaces when considering the termination of MMP: whenever a Mori fiber space appears, the MMP terminates.

2.2. **Homology groups.** We need to define two algebraic invariants, $\rho_c(X)$ and $\rho_r(X)$, for the definition of the difficulty function.

Definition 2.6 ([HLZ25, Definition 4.3])**.** Let X be a quasi-projective variety. We let $H_4^{\mathrm{alg}}(X, \mathbb{R})$ be the subspace in the singular homology group $H_4(X, \mathbb{R})$ generated by the classes of 2-dimensional complete algebraic subvarieties. We define $\rho_c(X) := \dim H_4^{\mathrm{alg}}(X, \mathbb{R})$.

In [HLZ25], $\rho_c(X)$ is denoted by $h_4^{\mathrm{alg}}(X, \mathbb{R})$. Since we only care about termination of flips in dimension 4, we will not use $h_{2m}^{\mathrm{alg}}(X, \mathbb{R})$ for any $m \neq 2$ in this paper.

Definition 2.7. Let X be a quasi-projective variety and $r \geq 0$ an integer. We let $H_{2r}^{BM,\mathrm{alg}}(X, \mathbb{Q})$ be the subspace in the Borel–Moore homology group $H_{2r}^{BM}(X, \mathbb{Q})$ ([Ful98, Section 19.1]) generated by the classes $[V] \in H_{2r}^{BM}(X, \mathbb{Q})$ of irreducible closed subvarieties of X of dimension r. We define $\rho_r(X) := \dim H_{2r}^{BM,\mathrm{alg}}(X, \mathbb{Q})$. It is clear that if X is projective, then

$$H_{2r}^{BM}(X, \mathbb{Q}) = H_{2r}(X, \mathbb{Q}).$$

We remark that we have to use the Borel–Moore homology group instead of the usual homology group $H_{2r}(X, \mathbb{Q})$ for our proof, as an irreducible closed subvariety of X which is not complete does not define a class in $H_{2r}(X, \mathbb{Q})$.

Lemma 2.8. *Let X be a quasi-projective variety and $r \geq 0$ an integer. Then $0 \leq \rho_c(X) < +\infty$ and $0 \leq \rho_r(X) < +\infty$.*

Proof. Dimension is always non-negative, so $0 \leq \rho_c(X)$ and $0 \leq \rho_r(X)$. Since $\dim H_{2r}^{BM}(X, \mathbb{Q}) < +\infty$ [Ful98, Example 19.1.3], $\rho_r(X) < +\infty$. Since X is quasi-projective, X is homotopy equivalent to a finite simplicial complex. To see this, we let $\overline{X}$ be a projective variety such that $X \subset \overline{X}$ is an open subset. By [Hir75], there exists a finite triangulation of $\overline{X}$ such that $\overline{X}\backslash X$ is a subcomplex and so X is homotopy equivalent to a finite simplicial complex. Thus $\rho_c(X) \leq \dim H_4(X, \mathbb{R}) < +\infty$. □

In the proof of the termination of flips, we need the monotonicity of $\rho_r(X)$ and $\rho_c(X)$. We need the following lemmas:

Lemma 2.9 ([HLZ25, Lemma 4.4])**.** *Let m be a non-negative integer and let $\pi : X \to Z$ be a birational morphism between normal quasi-projective varieties such that $\dim \mathrm{Ex}(\pi) \leq 2$ and $\mathrm{Ex}(\pi)$ has exactly m irreducible complete components of dimension 2. Then*

$$\rho_c(Z) + \min\{1, m\} \leq \rho_c(X) \leq \rho_c(Z) + m.$$

Lemma 2.10 (cf. [OAI26b, Lemma 2.3])**.** *Let X be a quasi-projective variety and $r \geq 0$ an integer. Let $V \subset X$ be a closed subset such that $\dim V \leq r$. Then the restriction map*

$$\tau : H_{2r}^{BM,\mathrm{alg}}(X, \mathbb{Q}) \to H_{2r}^{BM,\mathrm{alg}}(X\backslash V, \mathbb{Q})$$

is surjective, and $\ker(\tau) \subset H_{2r}^{BM}(X, \mathbb{Q})$ is spanned by the classes of r-dimensional irreducible components of V. In particular:

(1) $\rho_r(X) \geq \rho_r(X\backslash V)$.
(2) $\rho_r(X) = \rho_r(X\backslash V)$ if $\dim V < r$.

Proof. By [Ful98, Section 19.1, (6)], there exists an exact sequence

$$H_{2r}^{BM}(V, \mathbb{Q}) \xrightarrow{i} H_{2r}^{BM}(X, \mathbb{Q}) \xrightarrow{\tau} H_{2r}^{BM}(X\backslash V, \mathbb{Q})$$

where i is induced by the inclusion $V \to X$ and τ is induced by the restriction $X \to X\backslash V$. Now for any irreducible closed subvariety $W \subset X$ such that $\dim W = r$, we have that

$$\tau([W]) = [W\backslash V] \in H_{2r}^{BM,\mathrm{alg}}(X\backslash V, \mathbb{Q})$$

if $W \not\subset V$, and $\tau([W]) = 0$ if $W \subset V$. Since $\dim V \leq r$, the lemma follows. □

We remark that in Lemma 2.10(1), it is possible that $\rho_r(X) = \rho_r(X\backslash V)$ when $\dim V = r$. To obtain strict monotonicity of ρ_r in difficulty functions, we need the following lemma.

Lemma 2.11 (cf. [OAI26b, Lemma 7.1])**.** *Let $f : X \to Y$ be a birational morphism between normal quasi-projective threefolds. Let $E_1, \ldots, E_m$ be all f-exceptional prime divisors. Then*

$$[E_1], \ldots, [E_m] \in H_4^{BM}(X, \mathbb{Q})$$

are $\mathbb{Q}$-linearly independent.

Proof. Let $a_1, \ldots, a_m$ be rational numbers such that $\sum_{k=1}^m a_k[E_k] = 0 \in H_4^{BM}(X, \mathbb{Q})$. Let $X_{\mathrm{reg}} \subset X$ be the non-singular locus of X. Since X is normal, $\dim(X\backslash X_{\mathrm{reg}}) \leq 1$. By [KM98, Theorem 0.2], there exists a resolution $h : X' \to X$ such that h is an isomorphism over X_{reg}. We let $E_k' := h_*^{-1}E_k$ for any k and let $F_1, \ldots, F_n$ be the h-exceptional prime divisors.

Consider the restriction map

$$X \to X_{\mathrm{reg}} \cong X'\backslash \mathrm{Ex}(h)$$

which induces

$$H_4^{BM,\mathrm{alg}}(X, \mathbb{Q}) \to H_4^{BM,\mathrm{alg}}(X'\backslash \mathrm{Ex}(h), \mathbb{Q}) \leftarrow H_4^{BM,\mathrm{alg}}(X', \mathbb{Q}).$$

By Lemma 2.10, there exist rational numbers $b_1, \ldots, b_n$ such that

$$[D'] := \sum_{k=1}^m a_k[E_k'] + \sum_{j=1}^n b_j[F_j] = 0 \in H_4^{BM}(X', \mathbb{Q})$$

for $D' := \sum_{k=1}^{m} a_k E'_k + \sum_{j=1}^{n} b_j F_j$. Since X' is smooth, by [Ful98, Section 19.1, (3)] and [Ful98, Proposition 19.1.2], there exists an isomorphism

$$\mu : H^2(X', \mathbb{Q}) \to H_4^{BM}(X', \mathbb{Q})$$

such that $\mu(c_1(\mathcal{O}_{X'}(ND'))) = [ND']$, where N is a positive integer such that ND' is a Cartier divisor. Thus $ND' \equiv 0$, i.e. $D' \cdot C = 0$ for any complete curve $C \subset X'$. Since f is projective, $D' \equiv_Y 0$. By the negativity lemma [KM98, Lemma 3.39(1)], $D' = 0$. Thus $a_k = 0$ for any k, and the lemma follows. □

2.3. **Basic properties of flips.**

Definition 2.12. Let m and n be two positive integers. Let (X, B) be a pair and $\phi : X \xrightarrow{f} Z \xleftarrow{f^+} X^+$ a $(K_X + B)$-flip. We say that ϕ is an (m, n)-*flip* if $\dim \mathrm{Ex}(f) = m$ and $\dim \mathrm{Ex}(f^+) = n$.

Lemma 2.13 ([KMM87, Lemma 5-1-17], [AHK07, Lemma 1.2]). *Let d, m, n be three positive integers. Let $(X, B)/U$ be a klt pair of dimension d and $\phi : X \xrightarrow{f} Z \xleftarrow{f^+} X^+$ a $(K_X + B)$-flip/U. Assume that ϕ is an (m, n)-flip. Then $m + n \geq d - 1$. In particular, if $d = 4$, then*

$$(m, n) \in \{(2, 1), (2, 2), (1, 2)\}.$$

3. Basic reductions

In this section we reduce Theorem 1.3 to the case when (X, B) is $\mathbb{Q}$-factorial klt and only consider the termination of flips. Then we reduce Theorem 1.3 to the case when there exist infinitely many $(1, 2)$ or $(2, 2)$-flips.

Lemma 3.1 (cf. [Cas+25, Lemma 4.4]). *Let $(X, B)/U$ be an lc pair. Let*

$$\phi_i : X_i \xrightarrow{f_i} Z_i \xleftarrow{f_i^+} X_{i+1}, \quad X_0 := X$$

be a sequence of steps of a $(K_X + B)$-MMP/U, and let B_i be the image of B on X_i for each i. Let $h : X' \to X$ be a $\mathbb{Q}$-factorial dlt modification of (X, B) and let $K_{X'} + B' := h^*(K_X + B)$. Then there exists a sequence of birational maps/U

$$\phi'_i : X'_i \dashrightarrow X'_{i+1}, \quad X'_0 := X'$$

and birational morphisms $h_i : X'_i \to X_i$, $h_0 := h$ satisfying the following. Let B'_i be the image of B' on X'_i for each i. Then for each i:

(1) h_i is a $\mathbb{Q}$-factorial dlt modification of (X_i, B_i) and

$$K_{X'_i} + B'_i = h_i^*(K_{X_i} + B_i).$$

(2) $h_{i+1} \circ \phi'_i = \phi_i \circ h_i$.

(3) ϕ'_i is a $(K_{X'_i} + B'_i)$-MMP/Z_i that is not the identity morphism, and $(X'_{i+1}, B'_{i+1})/Z_i$ is a good minimal model of $(X'_i, B'_i)/Z_i$.

(4) $\{\phi'_i\}$ is a sequence of steps of a $(K_{X'} + B')$-MMP/U.

Proof. It is actually a special case of [Cas+25, Lemma 4.4] and some variations are well-known (cf. [Bir12, Remark 2.9]) but for the reader's convenience we provide a full proof here without introducing more complicated structures such as adjoint foliated structures.

(4) follows from (3) so we only need to prove (1)(2)(3). Let n be a non-negative integer. We prove the lemma by induction on n under the induction hypothesis that we have already constructed $(X'_i, B'_i)/U$ and h_i for any $i \leq n$ and ϕ'_i for any $i \leq n-1$ which satisfy (1)(2)(3). When $n = 0$, this follows from our assumption. We need to construct ϕ'_n, h_{n+1}, and $(X'_{n+1}, B'_{n+1})/U$.

Since the induced birational map $\phi_n \circ h_n : X'_n \dashrightarrow X_{n+1}$ is the ample model/Z_n of $K_{X'_n} + B'_n$, we may run a $(K_{X'_n} + B'_n)$-MMP/Z_n with scaling of an ample divisor A_n which terminates with a good minimal model $(X'_{n+1}, B'_{n+1})/Z_n$ of $(X'_n, B'_n)/Z_n$ associated with a birational map/Z_n $\phi'_n : X'_n \dashrightarrow X'_{n+1}$ and a birational morphism $h_{n+1} : X'_{n+1} \to X_{n+1}$. We let $B'_{n+1} := \phi'_{n,*} B'_n$.

Then ϕ'_n, h_{n+1} and $(X'_{n+1}, B'_{n+1})/U$ satisfy our requirement, and we are done by induction on n. □

Proposition 3.2. *Let d be a positive integer and assume Conjecture 1.1 in dimension $\leq d-1$. Let $(X,B)/U$ be an lc pair of dimension d and*

$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$

an infinite sequence of steps of a (K_X+B)-MMP/U. Then there exists a $\mathbb{Q}$-factorial klt pair $(Y, B_Y)/U$ of dimension d and an infinite sequence of steps of a (K_Y+B_Y)-MMP/U.

Proof. By Lemma 3.1, possibly replacing (X,B) with a $\mathbb{Q}$-factorial dlt model, we may assume that (X,B) is $\mathbb{Q}$-factorial dlt. By [Fuj07, Theorem 4.2.1], possibly truncating the MMP, we may assume that the MMP is an isomorphism near $\lfloor B_i \rfloor$ for any i. Possibly replacing B with $\{B\}$, we may assume that (X,B) is klt. The proposition follows. □

Proposition 3.3. *Let $(X,B)/U$ be an lc pair and let $\phi : X \xrightarrow{f} Z \xleftarrow{f^+} X^+$ be a (K_X+B)-flip/U with $B^+ := \phi_* B$. Let W be the normalization of the main component of $X \times_Z X^+$ associated with birational morphisms $p : W \to X$ and $q : W \to X^+$. Let*

$$E := p^*(K_X + B) - q^*(K_{X^+} + B^+).$$

Then the following hold.

(1) $0 \leq E$ is anti-ample/Z.
(2) $\operatorname{Supp} E = p^{-1}(\operatorname{Ex}(f)) = q^{-1}(\operatorname{Ex}(f^+))$.
(3) For any prime divisor F over X, $a(F, X, B) \leq a(F, X^+, B^+)$.
(4) For any prime divisor F over X,

$$a(F, X, B) < a(F, X^+, B^+) \Leftrightarrow \operatorname{center}_X F \subset \operatorname{Ex}(f) \Leftrightarrow \operatorname{center}_{X^+} F \subset \operatorname{Ex}(f^+).$$

Proof. Let A (resp. A^+) be any ample/Z $\mathbb{R}$-divisor on X (resp. X^+). We let $h : W \to Y$ (resp. $h^+ : W \to Y^+$) be the ample model/X^+ (resp. ample model/X) of p^*A (resp. q^*A^+). Then there exists a morphism $Y \to X^+$ (resp. $Y^+ \to X$) as h (resp. h^+) is a morphism over X^+ (resp. over X). Since A is ample, X (resp. X^+) is the ample model/Z of p^*A (resp. q^*A^+), so there exists an induced morphism $Y \to X$ (resp. $Y^+ \to X^+$). By the universal property of $X \times_Z X^+$ and Stein factorization and since h (resp. h^+) is a contraction, we have that h (resp. h^+) is an isomorphism.

In particular, $p^*(K_X+B)$ is anti-ample/X^+ and $q^*(K_{X^+}+B^+)$ is ample/X. Now for any $R \in \overline{NE}(W/Z)$, if $R \in \overline{NE}(W/X)$, then

$$E \cdot R = -q^*(K_{X^+} + B^+) \cdot R < 0,$$

and if $R \notin \overline{NE}(W/X)$, then

$$E \cdot R \leq p^*(K_X + B) \cdot R = (K_X + B) \cdot p_* R < 0.$$

By Kleiman's criterion, E is anti-ample/Z. Since ϕ is small, E is exceptional/X and exceptional/X^+. By applying the negativity lemma twice, $E \geq 0$. This implies (1). Since E is anti-ample/Z, by applying the negativity lemma again (cf. [KM98, Lemma 3.39(2)]), we get (2). (3) follows from (1) and (4) follows from (1) and (2). □

Proposition 3.4. *Let $(X,B)/U$ be a $\mathbb{Q}$-factorial klt pair of dimension 4 and let*

$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$

be a sequence of (K_X+B)-flips/U. Write $\phi_i : X_i \xrightarrow{f_i} Z_i \xleftarrow{f_i^+} X_{i+1}$ where f_i is the flipping contraction and f_i^+ is the flipped contraction. Then:

(1) For any i, if $\dim \operatorname{Ex}(f_i^+) \leq 1$, then $\rho_c(X_{i+1}) < \rho_c(X_i)$.
(2) If ϕ_i is a $(2,1)$-flip for any $i \gg 0$, then this sequence of flips terminates.

Proof. (1) Assume that $\dim \mathrm{Ex}(f_i^+) \leq 1$. By Lemma 2.13, ϕ_i is a $(2,1)$-flip, so $\dim \mathrm{Ex}(f_i) = 2$. Let $V_i := f_i(\mathrm{Ex}(f_i))$. Then $\dim V_i < \dim \mathrm{Ex}(f_i^+)$, so V_i is a finite set of closed points. Thus $\mathrm{Ex}(f_i) = f_i^{-1}(V_i)$ and $\mathrm{Ex}(f_i^+) = f_i^{+,-1}(V_i)$ are complete. By Lemma 2.9, $\rho_c(Z_i) = \rho_c(X_{i+1})$ and $\rho_c(Z_i) < \rho_c(X_i)$. This implies (1).

(2) Possibly truncating the MMP we may assume that ϕ_i is a $(2,1)$-flip for any i. By (1) and Lemma 2.8, the MMP terminates after $\leq \rho_c(X)$ steps. □

4. Lifting the MMP to terminal pairs

The goal of this section is to reduce Theorem 1.3 to the case when all non-terminal places of (X, B) are non-terminal places of (X_i, B_i) for any $i \gg 0$, and when their log discrepancies do not stabilize. We shall further reduce to the case when the centers of these non-terminal places are closed points. Then we construct an induced sequence of terminalizations of (X_i, B_i) which we shall use to construct difficulty functions later.

Definition-Lemma 4.1. Let $(X, B)/U$ be a $\mathbb{Q}$-factorial klt pair and let

$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$

be a sequence of $(K_X + B)$-flips$/U$. Let

$$\mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty}) := \{E \mid E \text{ is an exceptional}/X \text{ prime divisor}, a(E, X_i, B_i) \leq 1 \text{ for any } i \geq 0\}.$$

Then $\mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$ is a finite set. We also define

$$\mathcal{S}^{\mathrm{stab}}(X, B; \{\phi_i\}_{i=0}^{+\infty}) := \{E \in \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty}) \mid a(E, X_i, B_i) = a(E, X, B) \text{ for any } i \geq 0\}$$

and

$$\mathcal{S}^{\mathrm{mov}}(X, B; \{\phi_i\}_{i=0}^{+\infty}) := \{E \in \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty}) \mid a(E, X_i, B_i) < \lim_{j \to +\infty} a(E, X_j, B_j) \text{ for any } i \geq 0\}.$$

Proof. For any $E \in \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$, we have $a(E, X, B) \leq a(E, X_i, B_i) \leq 1$. Since (X, B) is klt, such E belongs to a finite set, so $\mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$ is a finite set. □

Definition 4.2. Let $(X, B)/U$ be a pair and let

$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$

be a sequence of $(K_X + B)$-flips$/U$. We say that $\{\phi_i\}_{i=0}^{+\infty}$ is a sequence of *sloppy* $(K_X + B)$*-flips*$/U$ if (X, B) is $\mathbb{Q}$-factorial klt and

$$\mathcal{S}^{\mathrm{mov}}(X, B; \{\phi_i\}_{i=0}^{+\infty}) = \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty}) = \{E \mid E \text{ is exceptional}/X, a(E, X, B) \leq 1\}.$$

We say that $\{\phi_i\}_{i=0}^{+\infty}$ is a sequence of *very sloppy* $(K_X + B)$*-flips*$/U$ if $\{\phi_i\}_{i=0}^{+\infty}$ is a sequence of sloppy $(K_X + B)$-flips$/U$ and $\mathrm{center}_{X_i} E$ is a closed point for any $E \in \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$ and any i.

Lemma 4.3. *Let d be a positive integer. Then the following conditions are equivalent:*

(1) For any $\mathbb{Q}$-factorial klt pair $(X, B)/U$ of dimension d, any sequence of $(K_X + B)$-flips$/U$ terminates.

(2) For any $\mathbb{Q}$-factorial klt pair $(X, B)/U$ of dimension d, any sequence of sloppy $(K_X + B)$-flips$/U$ terminates.

Proof. (1) implies (2) obviously. We only need to show that (2) implies (1). Assume that

$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$

is an infinite sequence of $(K_X + B)$-flips$/U$. For any $E \in \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$, one of the following conditions holds:

(1) $a(E, X_i, B_i) = \lim_{j \to +\infty} a(E, X_j, B_j)$ for any $i \gg 0$.
(2) $a(E, X_i, B_i) < \lim_{j \to +\infty} a(E, X_j, B_j)$ for any i.

Possibly truncating the MMP, we may assume that for any $E \in \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$, either $a(E, X, B) = a(E, X_j, B_j)$ for any j, or $a(E, X_i, B_i) < \lim_{j\to+\infty} a(E, X_j, B_j)$ for any i, i.e. either $E \in \mathcal{S}^{\mathrm{stab}}(X, B; \{\phi_i\}_{i=0}^{+\infty})$ or $E \in \mathcal{S}^{\mathrm{mov}}(X, B; \{\phi_i\}_{i=0}^{+\infty})$. Possibly truncating the MMP further, we may assume that for any non-terminal place E of (X, B) that is exceptional$/X$, E is also a non-terminal place of (X_i, B_i) for any i.

Since (X, B) is $\mathbb{Q}$-factorial klt, we may let $h : X' \to X$ be an extraction such that $\mathrm{Exc}(h)$ consists of exactly the prime divisors contained in $\mathcal{S}^{\mathrm{stab}}(X, B; \{\phi_i\}_{i=0}^{+\infty})$ and X' is $\mathbb{Q}$-factorial [BCHM10, Corollary 1.4.3]. We let $K_{X'} + B' := h^*(K_X + B)$. Write $\phi_i : X_i \xrightarrow{f_i} Z_i \xleftarrow{f_i^+} X_{i+1}$ where f_i is the flipping contraction and f_i^+ is the flipped contraction.

Claim 4.4. *There exists a sequence of birational maps$/U$*

$$\phi_i' : X_i' \dashrightarrow X_{i+1}', \quad X_0' := X'$$

and birational morphisms $h_i : X_i' \to X_i$, $h_0 := h$ satisfying the following. Let B_i' be the image of B' on X_i' for each i. Then for each i:

(1) We have
$$K_{X_i'} + B_i' = h_i^*(K_{X_i} + B_i).$$
(2) $h_{i+1} \circ \phi_i' = \phi_i \circ h_i$.
(3) ϕ_i' is a $(K_{X_i'} + B_i')$-MMP$/Z_i$ that is not the identity morphism, and $(X_{i+1}', B_{i+1}')/Z_i$ is a good minimal model of $(X_i', B_i')/Z_i$.
(4) ϕ_i' does not contract any divisor.
(5) $\{\phi_i'\}$ is an infinite sequence of $(K_{X'} + B')$-flips$/U$.

Proof. (5) follows from (3) and (4) so we only need to prove (1)(2)(3)(4). Let n be a non-negative integer. We prove the claim by induction on n under the induction hypothesis that we have already constructed $(X_i', B_i')/U$ and h_i for any $i \le n$ and ϕ_i' for any $i \le n-1$ which satisfy (1)(2)(3)(4). When $n = 0$, this follows from our assumption. We need to construct ϕ_n', h_{n+1}, and $(X_{n+1}', B_{n+1}')/U$.

Since the induced birational map $\phi_n \circ h_n : X_n' \dashrightarrow X_{n+1}$ is the ample model$/Z_n$ of $K_{X_n'} + B_n'$, we may run a $(K_{X_n'} + B_n')$-MMP$/Z_n$ with scaling of an ample divisor A_n which terminates with a good minimal model $(X_{n+1}', B_{n+1}')/Z_n$ of $(X_n', B_n')/Z_n$ associated with a birational map$/Z_n$ $\phi_n' : X_n' \dashrightarrow X_{n+1}'$ and a birational morphism $h_{n+1} : X_{n+1}' \to X_{n+1}$. We let $B_{n+1}' := \phi_{n,*}' B_n'$. Then ϕ_n', h_{n+1} and $(X_{n+1}', B_{n+1}')/U$ satisfy (1)(2)(3).

We are left to prove (4). Suppose that ϕ_n' contracts a prime divisor F. By induction, all exceptional$/Z_n$ prime divisors on X_n' are exactly the prime divisors contained in $\mathcal{S}^{\mathrm{stab}}(X, B; \{\phi_i\}_{i=0}^{+\infty})$. Since ϕ_n' is an MMP$/Z_n$, $F \in \mathcal{S}^{\mathrm{stab}}(X, B; \{\phi_i\}_{i=0}^{+\infty})$. Since F is contracted by ϕ_n', we have

$$a(F, X_n', B_n') < a(F, X_{n+1}', B_{n+1}').$$

This is not possible as

$$a(F, X_n', B_n') = a(F, X_n, B_n) = a(F, X_{n+1}, B_{n+1}) = a(F, X_{n+1}', B_{n+1}')$$

where the middle equality follows from $F \in \mathcal{S}^{\mathrm{stab}}(X, B; \{\phi_i\}_{i=0}^{+\infty})$. This implies (4) and the claim follows by induction on n. □

Proof of Lemma 4.3 continued. By Claim 4.4, there exists an infinite sequence of $(K_{X'} + B')$-flips$/U$

$$\phi_i' : (X_i', B_i') \dashrightarrow (X_{i+1}', B_{i+1}'), \quad (X_0', B_0') := (X', B'),$$

and birational morphisms $h_i : X_i' \to X_i$ such that

$$K_{X_i'} + B_i' = h_i^*(K_{X_i} + B_i)$$

for each i. By our construction,

$$\mathcal{S}^{\mathrm{mov}}(X', B'; \{\phi_i'\}_{i=0}^{+\infty}) = \mathcal{S}(X', B'; \{\phi_i'\}_{i=0}^{+\infty}) = \{E \mid E \text{ is exceptional}/X', a(E, X', B') \le 1\}.$$

By (2), this is not possible. The lemma follows. □

Lemma 4.5. *Let $(X, B)/U$ be a $\mathbb{Q}$-factorial klt pair of dimension 4,*

$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$

a sequence of sloppy $(K_X + B)$-flips/U, and $E \in \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$ a prime divisor. Then for any $i \gg 0$, $\operatorname{center}_{X_i} E$ is a closed point.

Proof. Suppose not. Then there exists an infinite sequence $0 \leq i_1 < i_2 < \cdots < i_k < \dots$ such that $V_{i_k} := \operatorname{center}_{X_{i_k}} E$ is not a closed point for any k. Since $\{\phi_i\}$ is sloppy,

$$a(E, X_{i_k}, B_{i_k}) < \lim_{j \to +\infty} a(E, X_{i_j}, B_{i_j})$$

for any k. Thus possibly passing to a subsequence, we may assume that

$$\{a(E, X_{i_k}, B_{i_k})\}_{k=1}^{+\infty}$$

is strictly increasing. Let H_{i_k} be the intersection of $\dim V_{i_k}$ general hyperplane sections $L_1, \dots, L_{\dim V_{i_k}}$ on X_{i_k} with x_{i_k} a closed point in $H_{i_k} \cap V_{i_k}$. Let

$$K_{H_{i_k}} + B_{H_{i_k}} := \left(K_{X_{i_k}} + B_{i_k} + \sum L_j\right)\Big|_{H_{i_k}}.$$

Then $\dim H_{i_k} \leq 3$ and there exist at most $\#\mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$ different log discrepancies in $[0, 1]$ of $(H_{i_k}, B_{H_{i_k}})$ over a neighborhood of x_{i_k}. By [HL25, Corollary 1.7], $a(E, X_{i_k}, B_{i_k})$ belongs to an ACC set, a contradiction. □

Lemma 4.6. *Let $(X, B)/U$ be a $\mathbb{Q}$-factorial klt pair of dimension 4 and*

$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$

a sequence of sloppy $(K_X + B)$-flips/U. Then for any $i \gg 0$ and any non-closed point $\eta_i \in X_i$, $(X_i \ni \eta_i, B_i)$ is terminal. In particular, X_i is smooth in codimension 2 for any $i \gg 0$.

Proof. By Lemma 4.5, $\operatorname{center}_{X_i} E$ is a closed point for any $E \in \mathcal{S}(X, B; \{\phi_i\}_{i=0}^{+\infty})$ and any $i \gg 0$. Thus for any $i \gg 0$ and any non-closed point $\eta_i \in X_i$, $(X_i \ni \eta_i, B_i)$ is terminal. Terminal implies smooth in codimension 2. □

Set-up 4.7. $(X, B), U, \phi_i, X_i, B_i, f_i, f_i^+, Z_i$ are as follows.

(1) $(X, B)/U$ is a $\mathbb{Q}$-factorial klt pair of dimension 4.
(2) The sequence
$$\phi_i : (X_i, B_i) \dashrightarrow (X_{i+1}, B_{i+1}), \quad (X_0, B_0) := (X, B)$$
is a sequence of very sloppy $(K_X + B)$-flips/U.
(3) We may write
$$\phi_i : X_i \xrightarrow{f_i} Z_i \xleftarrow{f_i^+} X_{i+1}$$
where f_i is the flipping contraction and f_i^+ is the flipped contraction.

Construction-Proposition 4.8. Notation and conditions as in Set-up 4.7. Then we may construct $h, X', B', m, S_k, n_i, b_{j,k}, h_i, \psi_i, \phi'_j, X'_j, S_{j,k}, B'_j, \Gamma_0, \Gamma$ in the following way.

(1) $h : X' \to X$ is a $\mathbb{Q}$-factorial terminalization of (X, B) with
$$K_{X'} + B' := h^*(K_X + B).$$
(2) $S_1, \dots, S_m$ are the h-exceptional prime divisors and the irreducible components of B.
(3) $0 = n_0 < n_1 < \cdots < n_i < \dots$ is a strictly increasing sequence of integers.
(4) $b_{n_i,k} = 1 - a(S_k, X_i, B_i)$ for any i, k.
(5) $h_i : X'_{n_i} \to X_i$ is a $\mathbb{Q}$-factorial terminalization of (X_i, B_i), $X'_0 = X'$, and $h_0 = h$.
(6) $K_{X'_{n_i}} + B'_{n_i} = h_i^*(K_{X_i} + B_i)$ for any i.

(7) We may write $B'_{n_i}=\sum_{k=1}^m b_{n_i,k}S_{n_i,k}$ where $S_{n_i,k}$ is the image of S_k on X'_{n_i} for any i,k.
(8) $\psi_i : X'_{n_i} \dashrightarrow X'_{n_{i+1}}$ is a birational map/Z_i such that $h_{i+1}\circ\psi_i=\phi_i\circ h_i$.
(9) ψ_i is a sequence of

$$\left(K_{X'_{n_i}}+\sum_{k=1}^m b_{n_{i+1},k}S_{n_i,k}\right)\text{-flips}/Z_i$$

and we denote by

$$\phi'_j : X'_j \dashrightarrow X'_{j+1},\quad n_i+1\le j\le n_{i+1}-1$$

this MMP, where $X'_{n_i+1}:=X'_{n_i}$.
(10) We let $S_{j,k}$ be the image of S_k on X'_j for any j,k, and let $b_{j,k}:=b_{n_{i+1},k}$ and $B'_j:=\sum_{k=1}^m b_{j,k}S_{j,k}$ for any $n_i+1\le j\le n_{i+1}$. Then (X'_j,B'_j) is a $\mathbb{Q}$-factorial terminal pair for any j.
(11) We denote by

$$\phi'_{n_i} : X'_{n_i}\to X'_{n_i+1}$$

the identity morphism.
(12) We let Γ_0 be the set of coefficients of B and let $\Gamma:=\{0\}\cup\{\sum l_i\gamma_i \mid \gamma_i\in\Gamma_0, l_i\in\mathbb{N}\}$.

Proof. (1) follows from [BCHM10, Corollary 1.4.3]. (2), (4), and (12) are definitions. (3) and (5-11) follow from [HLZ25, Lemma 2.8], Proposition 3.3(3), and the fact that $\{\phi_i\}$ is sloppy (hence ψ_i does not contract any divisor). □

5. The difficulty function

In this section we construct the difficulty function. Note that our difficulty function is different from the difficulty function defined in [OAI26b] as the latter could only handle rational coefficients. On the other hand, we note that, instead of considering a finite set of rational coefficients, we only need to consider a finitely generated set of coefficients which is closed under addition and define the weights in a similar way. We also remark that the definition of the weight function w_Γ does not follow the principle of the construction of the weight function as in [AHK07, Conditions 2.1] as those weights $w(\cdot)$ satisfy that $w(\cdot)$ is decreasing and $w(u)=0$ when $u>1$, while our weights are increasing.

5.1. Weights.

The reader may compare our weight with the weight defined in [OAI26b, Subsection 4.1].

Definition 5.1. A *finitely generated additive subset of* $\mathbb{R}_{\ge0}$ is of the form

$$\Gamma:=\{0\}\cup\left\{\sum_{j=1}^q m_jc_j \;\middle|\; m_1,\dots,m_q\in\mathbb{N}\right\}$$

where q is a non-negative integer and $c_1,\dots,c_q$ are positive real numbers.

Definition-Lemma 5.2. Let Γ be a finitely generated additive subset of $\mathbb{R}_{\ge0}$. For any real number $u\in\mathbb{R}$ and $b\in[0,1)$, we define

$$w_\Gamma(u):=\#\Gamma\cap[0,u)\quad\text{and}\quad d_\Gamma(b):=\sum_{k=1}^{+\infty}w_\Gamma(1-k(1-b)).$$

Then the following hold.

(1) $w_\Gamma:\mathbb{R}\to\mathbb{N}$ is an increasing function and $w_\Gamma(u)\ge1$ for any $u>0$.
(2) $d_\Gamma:[0,1)\to\mathbb{N}$ is an increasing function.
(3) $w_\Gamma(x+y)\ge w_\Gamma(x)+w_\Gamma(y)-1$ for any $x,y>0$.
(4) $w_\Gamma(x+y)\ge w_\Gamma(x)+w_\Gamma(y)$ for any $x\in\Gamma$ and any $y\ge0$.
(5) $w_\Gamma(x+y)\ge w_\Gamma(x)+1$ for any $x\in\Gamma$ and any $y>0$.

Proof. (1) w_Γ is obviously an increasing function. Since $0 \in \Gamma$, $w_\Gamma(u) \geq 1$ for any $u > 0$.

(2) For any $b \in [0,1)$, only finitely many values of $1-k(1-b)$ are non-negative, so $d_\Gamma(b)$ is well-defined. (2) follows from (1).

(3) Let $s_0 := \max \Gamma \cap [0,x)$. s_0 is well-defined as $0 \in \Gamma$. Since Γ is closed under addition and $s_0 + t < x + y$ for any $t \in [0,y)$, both $\Gamma \cap [0,x)$ and $s_0 + (\Gamma \cap (0,y))$ are subsets of $\Gamma \cap [0,x+y)$. Any element of $s_0 + (\Gamma \cap (0,y))$ is strictly larger than s_0, so $\Gamma \cap [0,x)$ and $s_0 + (\Gamma \cap (0,y))$ are disjoint. Since $\#(\Gamma \cap (0,y)) = w_\Gamma(y) - 1$, we have $w_\Gamma(x+y) \geq w_\Gamma(x) + w_\Gamma(y) - 1$.

(4) If $y = 0$, then $w_\Gamma(x+y) = w_\Gamma(x)$ and $w_\Gamma(y) = 0$ and (4) follows. Assume that $y > 0$. Then $\Gamma \cap [0,x)$ and $x + (\Gamma \cap [0,y))$ are disjoint subsets of $\Gamma \cap [0,x+y)$. We have $\#(\Gamma \cap [0,x)) = w_\Gamma(x)$ and $\#(x + (\Gamma \cap [0,y))) = w_\Gamma(y)$. This implies (4).

(5) It follows from (1) and (4). □

5.2. **Definition of difficulty.** In this subsection we define the difficulty function.

Lemma 5.3 (cf. [OAI26b, Lemma 3.3])**.** *Let $(X, B = \sum_{i=1}^m b_i S_i)$ be a $\mathbb{Q}$-factorial terminal pair of dimension 4 where S_i are the irreducible components of B. Then:*

(1) There exist only finitely many prime divisors E over X such that

$$\dim \operatorname{center}_X E \leq 1 \quad \text{and} \quad a(E,X,B) < 2.$$

(2) For any irreducible surface $T \subset X$, there exist only finitely many prime divisors E over X such that

$$\operatorname{center}_X E = T \quad \text{and} \quad a(E,X,B) < 2.$$

(3) For all but finitely many irreducible surfaces $T \subset X$ with generic point η_T we have the following:

(a) X is smooth near η_T.

(b) S_i is smooth near η_T for any i, and T is contained in at most one S_i.

(c) For any finitely generated additive subset Γ of $\mathbb{R}_{\geq 0}$, we have

$$\sum_{E \mid \operatorname{center}_X E = T} w_\Gamma(2 - a(E,X,B)) = \begin{cases} d_\Gamma(b_i) & \text{if } T \subset S_i, \\ 0 & \text{if } T \not\subset S_i \text{ for any } i \end{cases}.$$

Proof. (1)(2) follow from [AHK07, Lemma 1.5]. Since (X,B) is $\mathbb{Q}$-factorial terminal, X is smooth in codimension two, which implies (3.a). Let

$$Z := \operatorname{Sing} X \cup \bigcup_{i=1}^m \operatorname{Sing} S_i \cup \bigcup_{i \neq j} (S_i \cap S_j).$$

Then Z is a closed subset of X of codimension at least two, so Z contains only finitely many irreducible surfaces. Thus in (3) we may only consider irreducible surfaces $T \subset X$ such that $T \not\subset Z$, and we are left to prove (3.c). If $T \not\subset S_i$ for any i, then $a(E,X,B) = a(E,X,0) \geq 2$ for any prime divisor E over X such that $\operatorname{center}_X E = T$, and we have $w_\Gamma(2 - a(E,X,B)) = 0$. Thus we may assume that $T \subset S_i$ for some i. By [AHK07, Example 1.4],

$$\{E \mid \operatorname{center}_X E = T, a(E,X,B) < 2\} = \{E_1, \dots, E_n\}$$

where E_k is a prime divisor over X such that

$$\operatorname{center}_X E_k = T \quad \text{and} \quad a(E_k,X,B) = 1 + k(1-b_i),$$

and $n = \left\lceil \frac{1}{1-b_i} \right\rceil - 1$. We now have

$$\sum_{E \mid \operatorname{center}_X E = T} w_\Gamma(2 - a(E,X,B)) = \sum_{k=1}^n w_\Gamma(1 - k(1-b_i)) = d_\Gamma(b_i).$$

The lemma follows. □

Set-up 5.4. Notation and conditions as in Set-up 4.7 and Construction-Proposition 4.8. We further define:

(1) $d_{i,k} := d_\Gamma(b_{i,k})$ for any i, k.
(2) $\nu_{i,k} : S^\nu_{i,k} \to S_{i,k}$ is the normalization of $S_{i,k}$.
(3) For any irreducible surface $T \subset X'_i$ and any k, we denote by $r_{i,T,k}$ the number of prime divisors on $S^\nu_{i,k}$ whose image on X'_i is T.

We are ready to define difficulties. One may want to compare our definition with [AHK07, Definition 2.3]. They follow the same pattern despite the fact that the weight function w_Γ does not satisfy [AHK07, Conditions 2.1].

Definition-Proposition 5.5 (cf. [OAI26b, Definition 4.2])**.** Notation and conditions as in Set-up 4.7, Construction-Proposition 4.8, and Set-up 5.4. For any i, we define

$$\begin{aligned} D_i := & \sum_{E|\dim \operatorname{center}_{X'_i} E \le 1} w_\Gamma(2 - a(E, X'_i, B'_i)) \\ & + \sum_{T|T \text{ is an irreducible surface on } X'_i} \left(\sum_{E|\operatorname{center}_{X'_i} E = T} w_\Gamma(2 - a(E, X'_i, B'_i)) - \sum_{k=1}^m r_{i,T,k} d_{i,k} \right) \\ & + \sum_{k=1}^m d_{i,k} \rho_2(S^\nu_{i,k}). \end{aligned}$$

We say that D_i is the *difficulty* of (X'_i, B'_i). Then D_i is a well-defined integer. In particular, all the sums in the definition of D_i are finite sums.

Proof. We only need to show that D_i is well-defined and it will automatically be an integer.

By Lemma 5.3, the first sum is a finite sum. By Lemma 2.8, the last sum is a finite sum. By Lemma 5.3(3), for all but finitely many irreducible surfaces T on X'_i, one of the following holds.

- No $S_{i,k}$ contains T. Then $r_{i,T,k} = 0$ for any k, and the term corresponding to T in the middle term is 0 by Lemma 5.3(3).
- Exactly one $S_{i,k}$ contains T, and $S_{i,k}$ is smooth at η_T. Then $S^\nu_{i,k} \to S_{i,k}$ is an isomorphism over a neighborhood of η_T, so $r_{i,T,k} = 1$, and $r_{i,T,j} = 0$ for any $j \neq k$. By Lemma 5.3(3), the term corresponding to T in the middle term is 0.

Thus the middle term is also a finite sum. The Definition-Proposition follows. □

5.3. **Properties of the difficulty function.** In this subsection we study the monotonicity of the difficulty function. Compared with the classical difficulty function, it is much more difficult to show that the difficulty function is bounded and we shall leave its proof to Section 6.

Lemma 5.6. *Let X be a smooth surface, C a reduced curve on X, $x \in C$ a closed point, and $\nu : C^\nu \to C$ the normalization of C. Then*

$$\operatorname{mult}_x C \ge \#\nu^{-1}(x).$$

Proof. Let $\nu^{-1}(x) = \{x_1, \dots, x_n\}$. Locally analytically, we let $C_1, \dots, C_n$ be disjoint irreducible components of C^ν such that $x_i \in C_i$ for any i. Then

$$\operatorname{mult}_x C \ge \sum_{i=1}^n \operatorname{mult}_x \nu(C_i) \ge n = \#\nu^{-1}(x).$$

□

The proof of the following lemma is similar to [AHK07, Proof of Lemma 2.6].

Lemma 5.7. *Assume that:*

(1) Γ is a finitely generated additive subset of $\mathbb{R}_{\ge 0}$.

(2) $(X, B = \sum_{i=1}^n b_i S_i)$ *is a pair where* S_i *are the irreducible components of* B.
(3) X *is smooth and* $b_i \in [0, 1)$.
(4) T *is an irreducible subvariety of* X *such that* $\dim T = \dim X - 2$ *and* $T \subset S_i$ *for each* i.
(5) $\nu_i : S_i^\nu \to S_i$ *is the normalization of* S_i.
(6) $m_i := \mathrm{mult}_T S_i$.
(7) $\sum_{i=1}^n m_i b_i < 1$.
(8) $\{1, \dots, n\} = F \cup V$ *where* F, V *are disjoint, and* $b_i \in \Gamma$ *for any* $i \in F$.
(9) r_i *is the number of prime divisors on* S_i^ν *whose images on* S_i *are* T.
(10) $r := \sum_{i \in V} r_i$.

Then there exist finitely many distinct prime divisors $E_1, \dots, E_N$ *over* X, *such that* $\mathrm{center}_X E_j = T$ *for any* j *and*

$$\sum_{j=1}^N w_\Gamma(2 - a(E_j, X, B)) \geq \sum_{i=1}^n r_i d_\Gamma(b_i) - \max\{r - 1, 0\}.$$

Proof. We divide the proof into several steps.

Step 1. In this step we show that

$$w_\Gamma\Big(\sum_{i=1}^n r_i b_i\Big) \geq \sum_{i=1}^n r_i w_\Gamma(b_i) - \max\{r - 1, 0\}. \tag{5.1}$$

We define

$$u_F := \sum_{i \in F} r_i b_i \quad \text{and} \quad u_V := \sum_{i \in V} r_i b_i.$$

Since $b_i \in \Gamma$ for any $i \in F$ and Γ is closed under addition, $u_F \in \Gamma$. By Definition-Lemma 5.2(4), we have

$$w_\Gamma(u_F) \geq \sum_{i \in F} r_i w_\Gamma(b_i).$$

By Definition-Lemma 5.2(3), we have

$$w_\Gamma(u_V) \geq \sum_{i \in V} r_i w_\Gamma(b_i) - \max\{r - 1, 0\}.$$

Since $u_F \in \Gamma$, by Definition-Lemma 5.2(4) again, we have

$$w_\Gamma(u_F + u_V) \geq w_\Gamma(u_F) + w_\Gamma(u_V),$$

and we obtain (5.1).

Step 2. In this step we prove the lemma when $b_i \leq \frac{1}{2}$ for any i. Since X is smooth, by Lemma 5.6, we have $r_i \leq m_i$ for any i. Let $X' \to X$ be the ordinary blow-up of X at T with exceptional divisor E. Then

$$a(E, X, B) = 2 - \sum_{i=1}^n m_i b_i > 1.$$

By Definition-Lemma 5.2(1) and (5.1), we have

$$w_\Gamma(2 - a(E, X, B)) = w_\Gamma\Big(\sum_{i=1}^n m_i b_i\Big) \geq w_\Gamma\Big(\sum_{i=1}^n r_i b_i\Big) \geq \sum_{i=1}^n r_i w_\Gamma(b_i) - \max\{r - 1, 0\}. \tag{5.2}$$

Since $w_\Gamma(b) = d_\Gamma(b)$ for any $b \leq \frac{1}{2}$ by our definition, if $b_i \leq \frac{1}{2}$ for any i, then (5.2) implies the lemma by taking $N = 1$ and $E_1 = E$. Thus in the following, we may assume that $b_i > \frac{1}{2}$ for some i. Possibly reordering indices, we may assume that $b_1 > \frac{1}{2}$.

Step 3. In this step we prove the lemma. Since $b_1 > \frac{1}{2}$ and $m_1 b_1 < 1$, $m_1 = 1$, so $r_1 = 1$. Moreover, for any $i \geq 2$, we have $b_i < \frac{1}{2}$, so $w_\Gamma(b_i) = d_\Gamma(b_i)$ for any $i \geq 2$. Possibly shrinking X to a neighborhood of the generic point of T, we may assume that (X, S_1) is log smooth. By

[AHK07, Example 1.4], for any positive integer k such that $1+k(1-b_1)<2$, there exists a unique prime divisor E_k over X such that

$$\operatorname{center}_X E_k = T \quad \text{and} \quad a(E_k, X, B) \le a(E_k, X, b_1 S_1) = 1 + k(1-b_1),$$

and $E_1 = E$. By Definition-Lemma 5.2(1),

$$w_\Gamma(2 - a(E_k, X, B)) \ge w_\Gamma(1 - k(1-b_1)).$$

Let $N := \left\lceil \frac{1}{1-b_1} \right\rceil - 1$. Then we have

$$\begin{aligned}
\sum_{j=1}^{N} w_\Gamma(2 - a(E_j, X, B)) &\ge \sum_{i=1}^{n} r_i w_\Gamma(b_i) - \max\{r-1, 0\} + \sum_{k=2}^{N} w_\Gamma(1 - k(1-b_1)) \\
&= \sum_{k=1}^{N} w_\Gamma(1 - k(1-b_1)) + \sum_{i=2}^{n} r_i w_\Gamma(b_i) - \max\{r-1, 0\} \\
&= d_\Gamma(b_1) + \sum_{i=2}^{n} r_i d_\Gamma(b_i) - \max\{r-1, 0\} \\
&= \sum_{i=1}^{n} r_i d_\Gamma(b_i) - \max\{r-1, 0\}.
\end{aligned}$$

The lemma follows. □

Proposition 5.8. *Notation and conditions as in Set-up 4.7, Construction-Proposition 4.8, Set-up 5.4, and Definition-Proposition 5.5. Then there exists a positive integer N satisfying the following.*

(1) For any $i \ge N$ and any k, $d_{i,k} = d_{N,k}$.
(2) For any $j \ge i \ge N$, $D_i \ge D_j$.
(3) For any $j \ge i \ge N$ and any exceptional/X_i' prime divisor E, we have

$$D_i - D_j \ge w_\Gamma(2 - a(E, X_i', B_i')) - w_\Gamma(2 - a(E, X_j', B_j')).$$

(4) For any $j \ge i \ge N$, and any exceptional/X_i' prime divisor E, if

$$a(E, X_j', B_j') = 2 - t \quad \text{for some} \quad t \in \Gamma \quad \text{and} \quad a(E, X_i', B_i') < 2 - t,$$

then $D_i - D_j \ge 1$.

Proof. Since $b_{i,k}$ is decreasing for any fixed k, $d_{i,k}$ is a decreasing sequence of non-negative integers. This implies (1). Possibly truncating the MMP, we may assume that $d_{i,k}$ is a constant for any i. In the following we prove (2-4). We define

$$D_{i,T} := \sum_{E \mid \operatorname{center}_{X_i'} E = T} w_\Gamma(2 - a(E, X_i', B_i')) - \sum_{k=1}^{m} r_{i,T,k} d_{i,k}$$

for any i and any irreducible surface $T \subset X_i'$. It is clear that we only need to prove (2-4) for $j = i+1$. In the following, we fix i. We only need to show that

$$D_i - D_{i+1} = \sum_{E \mid E \text{ is exceptional}/X_i'} \Big(w_\Gamma(2 - a(E, X_i', B_i')) - w_\Gamma(2 - a(E, X_{i+1}', B_{i+1}')) \Big) \tag{5.3}$$

and then (2)(3) follow from Definition-Lemma 5.2(1), and (4) follows from Definition-Lemma 5.2(5).

We are left to prove (5.3). By our construction, one of the following cases holds:

(i) ϕ_i' is the identity morphism and $B_{i+1}' \le B_i'$.
(ii) ϕ_i' is a flip.

In Case (i), (5.3) is obvious as a direct consequence of Definition-Proposition 5.5. Thus we may assume that we are in Case (ii). In this case, $b_{i+1,k} = b_{i,k}$ for any k. Since ϕ'_i is a flip, we may write

$$\phi'_i : X'_i \xrightarrow{f'_i} Z'_i \xleftarrow{f'^{+}_i} X'_{i+1}$$

where f'_i is the flipping contraction and f'^{+}_i is the flipped contraction. Let

$$\mathcal{E} := \{E \mid E \text{ is exceptional}/X'_i, \operatorname{center}_{X'_i} E \subset \operatorname{Ex}(f'_i)\}.$$

Then

$$\mathcal{E} = \{E \mid E \text{ is exceptional}/X'_i, \operatorname{center}_{X'_{i+1}} E \subset \operatorname{Ex}(f'^{+}_i)\}.$$

We define

$$D^\circ_i := \sum_{E\notin\mathcal{E}, \dim \operatorname{center}_{X'_i} E\leq 1} w_\Gamma(2 - a(E, X'_i, B'_i)) + \sum_{T|T \text{ is an irreducible surface on } X'_i, T\not\subset \operatorname{Ex}(f'_i)} D_{i,T}$$

and

$$D^\circ_{i+1} := \sum_{E\notin\mathcal{E}, \dim \operatorname{center}_{X'_{i+1}} E\leq 1} w_\Gamma(2-a(E, X'_{i+1}, B'_{i+1}))+ \sum_{T|T \text{ is an irreducible surface on } X'_{i+1}, T\not\subset \operatorname{Ex}(f'^{+}_i)} D_{i+1,T}.$$

Then D°_i and D°_{i+1} are well-defined by Definition-Proposition 5.5. Moreover, we have $D^\circ_i = D^\circ_{i+1}$ by considering the definition of each term and using the fact that $b_{i+1,k} = b_{i,k}$ which implies that $d_{i+1,k} = d_{i,k}$.

We now define

$$e^-_{i,k} := \#\{E \mid E \text{ is a prime divisor on } S^\nu_{i,k} \text{ and } \nu_{i,k}(E) \subset \operatorname{Ex}(f'_i)\}$$

and

$$e^+_{i,k} := \#\{E \mid E \text{ is a prime divisor on } S^\nu_{i+1,k} \text{ and } \nu_{i+1,k}(E) \subset \operatorname{Ex}(f'^{+}_i)\}.$$

Then we have

$$e^-_{i,k} = \sum_{T\subset \operatorname{Ex}(f'_i)} r_{i,T,k} \quad \text{and} \quad e^+_{i,k} = \sum_{T\subset \operatorname{Ex}(f'^{+}_i)} r_{i+1,T,k}.$$

Since $\dim \operatorname{Ex}(f'_i) \leq 2$ and $\dim \operatorname{Ex}(f'^{+}_i) \leq 2$, there exist only finitely many irreducible surfaces in $\operatorname{Ex}(f'_i)$ and $\operatorname{Ex}(f'^{+}_i)$. By Definition-Proposition 5.5, we have

$$D_i = D^\circ_i + \sum_{E\in\mathcal{E}} w_\Gamma(2 - a(E, X'_i, B'_i)) - \sum_{k=1}^{m} d_{i,k}e^-_{i,k} + \sum_{k=1}^{m} d_{i,k}\rho_2(S^\nu_{i,k}), \tag{5.4}$$

$$D_{i+1} = D^\circ_{i+1} + \sum_{E\in\mathcal{E}} w_\Gamma(2 - a(E, X'_{i+1}, B'_{i+1})) - \sum_{k=1}^{m} d_{i+1,k}e^+_{i,k} + \sum_{k=1}^{m} d_{i+1,k}\rho_2(S^\nu_{i+1,k}). \tag{5.5}$$

Since $d_{i+1,k} = d_{i,k}$ and $D^\circ_i = D^\circ_{i+1}$, we have

$$\begin{aligned} D_i - D_{i+1} = &\sum_{E\in\mathcal{E}} \Big(w_\Gamma(2 - a(E, X'_i, B'_i)) - w_\Gamma(2 - a(E, X'_{i+1}, B'_{i+1}))\Big) \\ &+ \sum_{k=1}^{m} d_{i,k} \left(\left(\rho_2(S^\nu_{i,k}) - \rho_2(S^\nu_{i+1,k})\right) - \left(e^-_{i,k} - e^+_{i,k}\right)\right). \end{aligned}$$

Since

$$w_\Gamma(2 - a(E, X'_i, B'_i)) = w_\Gamma(2 - a(E, X'_{i+1}, B'_{i+1}))$$

for any $E \notin \mathcal{E}$, we only need to show that

$$\rho_2(S^\nu_{i,k}) - \rho_2(S^\nu_{i+1,k}) = e^-_{i,k} - e^+_{i,k} \tag{5.6}$$

for any k. We let $\tau_{i,k} : S^\nu_{i,k} \to X'_i$ and $\tau_{i+1,k} : S^\nu_{i+1,k} \to X'_{i+1}$ be the induced morphisms. Since f'_i, f'^+_i are small, the induced morphisms

$$S^\nu_{i,k} \to f'_i(S_{i,k}), \quad S^\nu_{i+1,k} \to f'^+_i(S_{i+1,k})$$

are birational. We let

$$S^\nu_{i,k} \xrightarrow{\alpha_i} W_{i,k} \to f'_i(S_{i,k}), \quad S^\nu_{i+1,k} \xrightarrow{\alpha^+_i} W^+_{i,k} \to f'^+_i(S_{i+1,k})$$

be the Stein factorizations. Then $W_{i,k} \to f'_i(S_{i,k})$ and $W^+_{i,k} \to f'^+_i(S_{i+1,k})$ are normalizations. By the definition of $e^-_{i,k}$ and $e^+_{i,k}$, the prime divisors contracted by α_i (resp. α^+_i) are exactly the $e^-_{i,k}$ (resp. $e^+_{i,k}$) prime divisors on $S^\nu_{i,k}$ (resp. $S^\nu_{i+1,k}$) whose images on X'_i (resp. X'_{i+1}) are contained in $\mathrm{Ex}(f'_i)$ (resp. $\mathrm{Ex}(f'^+_i)$). By Lemma 2.10, the kernel of the restriction

$$H_4^{BM,\mathrm{alg}}(S^\nu_{i,k}, \mathbb{Q}) \to H_4^{BM,\mathrm{alg}}(S^\nu_{i,k} \backslash \tau^{-1}_{i,k}(\mathrm{Ex}(f'_i)), \mathbb{Q})$$

is spanned by the classes of the $e^-_{i,k}$ prime divisors contracted by α_i, and the kernel of the restriction

$$H_4^{BM,\mathrm{alg}}(S^\nu_{i+1,k}, \mathbb{Q}) \to H_4^{BM,\mathrm{alg}}(S^\nu_{i+1,k} \backslash \tau^{-1}_{i+1,k}(\mathrm{Ex}(f'^+_i)), \mathbb{Q})$$

is spanned by the classes of the $e^+_{i,k}$ prime divisors contracted by α^+_i. By Lemma 2.11, we have

$$\rho_2(S^\nu_{i,k}) = \rho_2(S^\nu_{i,k} \backslash \tau^{-1}_{i,k}(\mathrm{Ex}(f'_i))) + e^-_{i,k}$$

and

$$\rho_2(S^\nu_{i+1,k}) = \rho_2(S^\nu_{i+1,k} \backslash \tau^{-1}_{i+1,k}(\mathrm{Ex}(f'^+_i))) + e^+_{i,k}.$$

Since

$$S^\nu_{i+1,k} \backslash \tau^{-1}_{i+1,k}(\mathrm{Ex}(f'^+_i)) \cong S^\nu_{i,k} \backslash \tau^{-1}_{i,k}(\mathrm{Ex}(f'_i)),$$

we obtain (5.6), and the proposition follows. □

6. Boundedness of difficulty

The goal of this section is to prove the following proposition.

Proposition 6.1. *Notation and conditions as in Set-up 4.7, Construction-Proposition 4.8, Set-up 5.4, and Definition-Proposition 5.5. Then $D_i \geq 0$ for any $i \gg 0$.*

The key ingredient of the proof is Theorem 6.13, which relies on Hodge theory.

6.1. Prime divisors over singular curves.

In this subsection, we construct prime divisors of log discrepancy 2 over the singular curves of a terminal fourfold.

Definition 6.2. Let X be a normal quasi-projective fourfold. We denote by IC_X the intersection complex of X such that $\mathrm{IC}_X|_{X_{\mathrm{reg}}} = \mathbb{Q}_{X_{\mathrm{reg}}}[4]$. It is a pure Hodge module of weight 4 [Sai89, 1.13]. In the rest of this section, we use the derived categories of mixed Hodge modules [Sai89, Theorems 1.3 and 1.4], and we do not distinguish a complex of mixed Hodge modules from the underlying complex of $\mathbb{Q}$-sheaves in the analytic topology.

Lemma 6.3. *Let X be a normal quasi-projective fourfold and X_{reg} the non-singular locus of X associated with inclusion $j : X_{\mathrm{reg}} \to X$. Then:*

1. *$\mathcal{H}^q(\mathrm{IC}_X)_x = 0$ for any closed point $x \in X$ and any $q \geq 0$.*
2. *Assume that $\dim \mathrm{Sing}\, X \leq 1$. Then for any irreducible curve $C \subset \mathrm{Sing}\, X$, there exists a dense open subset $C^\circ \subset C$ such that $\mathcal{H}^q(\mathrm{IC}_X)|_{C^\circ} = 0$ for any $q \geq -1$, and the morphism $\mathrm{IC}_X \to Rj_*\mathbb{Q}_{X_{\mathrm{reg}}}[4]$, which is the identity morphism on X_{reg}, induces an isomorphism*

$$\mathcal{H}^{-2}(\mathrm{IC}_X)|_{C^\circ} \to (R^2 j_* \mathbb{Q}_{X_{\mathrm{reg}}})|_{C^\circ}.$$

Proof. It follows from the truncation formula for IC_X [GM83, Sections 2.3, 3.1, and 3.3]. □

Lemma 6.4 (cf. [OAI26b, Lemma 6.5]). *Let $f : X \to Z$ be a projective morphism between quasi-projective schemes, $z \in Z$ a (not necessarily closed) point, and X_z the scheme-theoretic fiber of f over z. Assume that*

$$\dim X_z \leq 2 \quad \text{and} \quad H^2(X_z, \mathcal{O}_{X_z}) = 0.$$

Let $\mathcal{L}_1, \dots, \mathcal{L}_m$ be line bundles on X_z. Then there exist an étale morphism $q : Z' \to Z$ and a point $z' \in Z'$ satisfying the following.

(1) $q(z') = z$.
(2) q induces an isomorphism between $k(z')$ and $k(z)$.
(3) $\mathcal{L}_i = \mathcal{L}_i'|_{X'_{z'}}$ for some line bundle $\mathcal{L}_i'$, where $X' := X \times_Z Z'$ and $X'_{z'}$ is the fiber of $X' \to Z'$ over z'.

Proof. Taking fiber products of étale morphisms over Z, we may assume that $m = 1$ and we let $\mathcal{L} := \mathcal{L}_1$. Let $R := \mathcal{O}_{Z,z}$ with maximal ideal $\mathfrak{m}$ and let $X_n := X \times_Z \operatorname{Spec}(R/\mathfrak{m}^{n+1})$ be the thickenings for any $n \geq 0$. The ideal $\mathcal{J}_n$ of X_n in X_{n+1} is a quotient of $\mathcal{O}_{X_0}^{\oplus c}$ for some positive integer c. Since $\dim X_0 \leq 2$ and $H^2(X_0, \mathcal{O}_{X_0}) = 0$, we have

$$H^2(X_0, \mathcal{J}_n) = 0.$$

By the exact sequence

$$0 \to \mathcal{J}_n \to \mathcal{O}_{X_{n+1}}^* \to \mathcal{O}_{X_n}^* \to 1,$$

any line bundle on X_n extends to X_{n+1}. Thus $\mathcal{L}$ extends to a compatible system of line bundles on X_n, which extends to a line bundle on $X \times_Z \operatorname{Spec} \widehat{R}$ by Grothendieck's existence theorem [Gro61a, Théorème 5.1.4 and Scholie 5.1.7]. The functor which sends a Z-scheme Z' to the set of isomorphism classes of line bundles on $X \times_Z Z'$ is locally of finite presentation, and the lemma follows from Artin approximation [Art69, Corollary 2.2]. □

Lemma 6.5 (cf. [OAI26b, Lemma 6.7]). *Let $\pi : X' \to X$ be an étale morphism between normal $\mathbb{Q}$-Gorenstein quasi-projective varieties. Let $C \subset X$ and $C' \subset X'$ be irreducible curves such that $\pi|_{C'} : C' \to C$ is birational. Let $E'_1, \dots, E'_m$ be distinct prime divisors over X' such that $\operatorname{center}_{X'} E'_i = C'$ for any $1 \leq i \leq m$.*

Then there exist distinct prime divisors $E_1, \dots, E_m$ over X such that $\operatorname{center}_X E_i = C$ and

$$a(E_i, X) = a(E'_i, X')$$

for any i.

Proof. Since $\pi|_{C'}$ is birational, possibly shrinking X around the generic point of C and shrinking X' around the generic point of C', we may assume that π induces an isomorphism $C' \to C$ and $\pi^{-1}(C) = C'$. For any birational morphism $h : Y \to X$, we let $Y' := Y \times_X X'$ associated with the induced morphisms $\pi' : Y' \to Y$ and $h' : Y' \to X'$. Then π' is étale and induces an isomorphism $h'^{-1}(C') \to h^{-1}(C)$. Thus taking preimages under π' gives a bijection from the set of prime divisors D on Y such that $h(D) = C$ to the set of prime divisors D' on Y' such that $h'(D') = C'$, and $a(\pi'^{-1}(D), X') = a(D, X)$ for any such D by [KM98, Proposition 5.20].

By [KM98, Lemma 2.45], for any i, E'_i is a prime divisor on the variety obtained from X' by successively blowing up the centers of E'_i and normalizing, and these centers lie over C'. Since blow-ups and normalizations commute with étale base change, by the isomorphisms above and induction on the number of blow-ups, for any i there exists a birational morphism $h_i : Y_i \to X$ such that E'_i is a prime divisor on $Y_i \times_X X'$. Let $h : Y \to X$ be a birational morphism which factors through h_i for any i. Then $E'_1, \dots, E'_m$ are prime divisors on Y', and the lemma follows by letting $E_i := \pi'(E'_i)$ for any i. □

Definition-Lemma 6.6. Let X be a terminal quasi-projective fourfold and X_{reg} the non-singular locus of X associated with inclusion $j : X_{\mathrm{reg}} \to X$. Let $C \subset \operatorname{Sing} X$ be an irreducible curve. Let $C^\circ \subset C$ be a connected smooth dense open subset such that

$$\mathbb{V}_C := (R^2 j_* \mathbb{Q}_{X_{\mathrm{reg}}})|_{C^\circ}$$

is a local system. We define

$$\beta_C := \dim H^2_c(C^\circ, \mathbb{V}_C).$$

Then β_C is the dimension of the coinvariants of a fiber of $\mathbb{V}_C$ under the monodromy, and β_C does not depend on the choice of C°.

Proof. It follows from Poincaré duality. □

Proposition 6.7 (cf. [OAI26b, Proposition 6.1])**.** *Let X be a terminal quasi-projective fourfold and X_{reg} the non-singular locus of X associated with inclusion $j : X_{\mathrm{reg}} \to X$. Let $C \subset \operatorname{Sing} X$ be an irreducible curve. Then there exist at least β_C distinct prime divisors E over X such that* $\operatorname{center}_X E = C$ *and* $a(E, X, 0) = 2$.

Proof. We divide the proof into several steps. By [KM98, Theorem 0.2], there exists a resolution $h : W \to X$ such that h is an isomorphism over X_{reg}. Since X is terminal and $\dim X = 4$, $\dim \operatorname{Sing} X \leq 1$ and $\dim h^{-1}(C) \leq 3$. By Lemma 6.3(2), there exists a smooth dense open subset $C^\circ \subset C$ such that

- $\dim h^{-1}(y) \leq 2$ for any $y \in C^\circ$,
- $\mathbb{V}_C := (R^2 j_* \mathbb{Q}_{X_{\mathrm{reg}}})|_{C^\circ}$ and $\mathbb{W} := (R^2 h_* \mathbb{Q}_W)|_{C^\circ}$ are local systems, and
- the morphism $\mathrm{IC}_X \to R j_* \mathbb{Q}_{X_{\mathrm{reg}}}[4]$, which is the identity morphism on X_{reg}, induces an isomorphism
$$\mathcal{H}^{-2}(\mathrm{IC}_X)|_{C^\circ} \to (R^2 j_* \mathbb{Q}_{X_{\mathrm{reg}}})|_{C^\circ}.$$

We let $F := W \times_X C^\circ$, and we let $\tau : \mathbb{W} \to \mathbb{V}_C$ be the morphism induced by the restriction

$$R h_* \mathbb{Q}_W \to R j_* \mathbb{Q}_{X_{\mathrm{reg}}}.$$

For any (not necessarily closed) point $y \in C^\circ$, we denote by F_y the fiber of $F \to C^\circ$ over y.

Step 1. In this step we show that $H^2(F_y, \mathcal{O}_{F_y}) = 0$ for any $y \in C^\circ$ and that τ is surjective.

Since X is terminal, $R^2 h_* \mathcal{O}_W = 0$. Let $y \in C^\circ$, let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{X,y}$, and let $F_{y,n} := W \times_X \operatorname{Spec}(\mathcal{O}_{X,y}/\mathfrak{m}^{n+1})$ for any $n \geq 0$. Since $\dim F_{y,n} \leq 2$, the map

$$H^2(F_{y,n+1}, \mathcal{O}_{F_{y,n+1}}) \to H^2(F_{y,n}, \mathcal{O}_{F_{y,n}})$$

is surjective for any n. By the theorem on formal functions [Har77, Theorem III.11.1], $\varprojlim_n H^2(F_{y,n}, \mathcal{O}_{F_{y,n}}) = 0$, so $H^2(F_y, \mathcal{O}_{F_y}) = 0$.

By the decomposition theorem [Sai89, Corollary 1.8, Proposition 1.9, and Theorems 1.12 and 2.3], there exists a morphism $\mathrm{IC}_X \to R h_* \mathbb{Q}_W[4]$ which is the identity on X_{reg}. The composition

$$\mathrm{IC}_X \to R h_* \mathbb{Q}_W[4] \to R j_* \mathbb{Q}_{X_{\mathrm{reg}}}[4]$$

is the identity morphism on X_{reg} and induces the isomorphism

$$\mathcal{H}^{-2}(\mathrm{IC}_X)|_{C^\circ} \to (R^2 j_* \mathbb{Q}_{X_{\mathrm{reg}}})|_{C^\circ} = \mathbb{V}_C,$$

which factors through

$$\tau : \mathbb{W} = (R^2 h_* \mathbb{Q}_W)|_{C^\circ} \to (R^2 j_* \mathbb{Q}_{X_{\mathrm{reg}}})|_{C^\circ} = \mathbb{V}_C.$$

Thus τ is surjective.

Step 2. In this step we show that, possibly shrinking C°, there exist line bundles

$$\mathcal{L}_1, \dots, \mathcal{L}_{\beta_C}$$

on F, such that for any closed point $x \in C^\circ$, the classes

$$\tau(c_1(\mathcal{L}_i|_{F_x})), \quad 1 \leq i \leq \beta_C$$

are linearly independent in $\mathbb{V}_{C,x}$, where $\mathbb{V}_{C,x}$ is the fiber of $\mathbb{V}_C$ over x.

Let $x \in C^\circ$ be a closed point. By **Step 1**, the exponential sequence, and GAGA [Har77, Appendix B, Theorem 2.1], the map $c_1 : \operatorname{Pic}(F_x) \to H^2(F_x, \mathbb{Z})$ is surjective. By proper base

change, the fiber $\mathbb{W}_x$ of $\mathbb{W}$ over x is $H^2(F_x,\mathbb{Q})$, so there exist line bundles $\mathcal{M}_1,\dots,\mathcal{M}_n$ on F_x so that $c_1(\mathcal{M}_i)$ span $\mathbb{W}_x$.

By **Step 1** and Lemma 6.4, there exist an irreducible smooth curve $\widetilde{C}$, an étale morphism $\widetilde{C}\to C^\circ$, and a closed point $\widetilde{x}\in\widetilde{C}$ over x such that $\mathcal{M}_1,\dots,\mathcal{M}_n$ are the restrictions of line bundles $\widetilde{\mathcal{M}}_1,\dots,\widetilde{\mathcal{M}}_n$ on $\widetilde{F}:=F\times_{C^\circ}\widetilde{C}$ to the fiber $\widetilde{F}_{\widetilde{x}}\cong F_x$ over $\widetilde{x}$. Possibly shrinking C° and $\widetilde{C}$, and possibly replacing x and $\widetilde{x}$, we may assume that $\widetilde{C}\to C^\circ$ is finite of degree r for some positive integer r. Since $c_1(\mathcal{M}_i)$ span $\mathbb{W}_x$, $c_1\left(\widetilde{\mathcal{M}}_i\right)$ span the fiber over any closed point of $\widetilde{C}$.

Let $G:=\pi_1(C^\circ,x)$. The subgroup $\pi_1\left(\widetilde{C},\widetilde{x}\right)\subset G$ has finite index and fixes $c_1(\widetilde{\mathcal{M}}_i|_{\widetilde{F}_{\widetilde{x}}})$ for any i, so $\pi_1\left(\widetilde{C},\widetilde{x}\right)$ acts trivially on $\mathbb{W}_x$. Thus G acts on $\mathbb{W}_x$ through a finite group. Since τ is surjective, G acts on $\mathbb{V}_{C,x}$ through a finite group, $\mathbb{W}_x^G\to\mathbb{V}_{C,x}^G$ is surjective, and $\dim\mathbb{V}_{C,x}^G=\beta_C$ by Definition-Lemma 6.6.

Let $u\in\mathbb{W}_x^G$. Possibly replacing u with a positive multiple, we may write

$$u=\sum_{i=1}^n a_ic_1\left(\widetilde{\mathcal{M}}_i|_{\widetilde{F}_{\widetilde{x}}}\right)$$

with $a_i\in\mathbb{Z}$. We let

$$\widetilde{\mathcal{M}}:=\bigotimes_{i=1}^n\widetilde{\mathcal{M}}_i^{\otimes a_i},$$

and let $\mathcal{L}$ be the norm of $\widetilde{\mathcal{M}}$ along the finite étale morphism $\widetilde{F}\to F$ [Gro61, (6.5.5) and Proposition 6.5.8]. Since $\widetilde{C}\to C^\circ$ is of degree r, we have

$$\mathcal{L}|_{F_x}\cong\bigotimes_{k=1}^r\widetilde{\mathcal{M}}|_{\widetilde{F}_{\widetilde{x}_k}},$$

where $\widetilde{x}_1,\dots,\widetilde{x}_r$ are the points of $\widetilde{C}$ over x and $\widetilde{F}_{\widetilde{x}_k}\cong F_x$. Since $\widetilde{C}$ is connected and u is fixed by G, $c_1(\widetilde{\mathcal{M}}|_{\widetilde{F}_{\widetilde{x}_k}})=u$ for any k, so $c_1(\mathcal{L}|_{F_x})=ru$. We conclude this step as τ is surjective, u is arbitrary, $\dim\mathbb{V}_{C,x}^G=\beta_C$, and $\mathbb{V}_C$ is a local system, so linear independence at x gives linear independence at an arbitrary closed point of C°.

Step 3. In this step we extend $\mathcal{L}_1,\dots,\mathcal{L}_{\beta_C}$ to an étale neighborhood of the generic point of C° and construct a small $\mathbb{Q}$-factorialization of the neighborhood.

Let η be the generic point of C°. By **Step 1** and Lemma 6.4, there exist an étale morphism $q:\widetilde{X}\to X$ and a point $\widetilde{\eta}\in\widetilde{X}$ over η such that q induces an isomorphism between the residue fields of $\widetilde{\eta}$ and η, and $\mathcal{L}_1|_{F_\eta},\dots,\mathcal{L}_{\beta_C}|_{F_\eta}$ are the restrictions of line bundles $\widetilde{\mathcal{L}}_1,\dots,\widetilde{\mathcal{L}}_{\beta_C}$ on $\widetilde{W}:=W\times_X\widetilde{X}$ to the fiber $\widetilde{W}_{\widetilde{\eta}}\cong F_\eta$ over $\widetilde{\eta}$. We let C' be the closure of $\widetilde{\eta}$ in $\widetilde{X}$, and we let $\widetilde{h}:\widetilde{W}\to\widetilde{X}$ be the projection. Possibly shrinking C° and $\widetilde{X}$, we may assume that

- $\widetilde{X}$ is affine and irreducible and $\operatorname{Sing}\widetilde{X}=C'$,
- q induces an isomorphism $C'\to C^\circ$, and
- for any i and any closed point $x'\in C'$, we have $\widetilde{\mathcal{L}}_i|_{\widetilde{W}_{x'}}\cong\mathcal{L}_i|_{F_{q(x')}}$, where $\widetilde{W}_{x'}\cong F_{q(x')}$ is the fiber of $\widetilde{h}$ over x'.

Then $\widetilde{W}$ is smooth and irreducible, and $\widetilde{h}$ is an isomorphism over $\widetilde{X}\backslash C'$. For any i, we let $\widetilde{L}_i$ be a divisor on $\widetilde{W}$ such that $\mathcal{O}_{\widetilde{W}}(\widetilde{L}_i)\cong\widetilde{\mathcal{L}}_i$.

Since q is étale, $\widetilde{X}$ is terminal. Let $g:X'\to\widetilde{X}$ be a small $\mathbb{Q}$-factorialization of $\widetilde{X}$. Then X' is terminal and g is an isomorphism over $\widetilde{X}\backslash C'$. Let $T_1,\dots,T_m$ be the irreducible surfaces in X' such that $g(T_k)=C'$ for any k, and let T_k^ν be the normalization of T_k for any k. Possibly shrinking C° and $\widetilde{X}$ around $\widetilde{\eta}$, we may assume that $\operatorname{Ex}(g)=\bigcup_{k=1}^m T_k$, and that for any k, the

morphism $T_k^\nu \to Z_k$ in the Stein factorization $T_k^\nu \to Z_k \to C'$ has irreducible fibers. We let γ_k be the image on X' of a general fiber of $T_k^\nu \to Z_k$.

Step 4. In this step we show that $m \geq \beta_C$. For any i, we let $L_i' := g_*^{-1}(\widetilde{h}_*\widetilde{L}_i)$. We only need to show that the vectors

$$((L_i' \cdot \gamma_1), \dots, (L_i' \cdot \gamma_m)) \in \mathbb{Q}^m, \quad 1 \leq i \leq \beta_C$$

are linearly independent.

Let $\lambda_1, \dots, \lambda_{\beta_C}$ be integers such that $L' := \sum_i \lambda_i L_i'$ satisfies $(L' \cdot \gamma_k) = 0$ for any k. Then $L' \equiv_{\widetilde{X}} 0$. By the base-point-freeness theorem, $L' \sim_{\mathbb{Q},\widetilde{X}} 0$. Thus

$$NL' \sim g^*M$$

for some positive integer N and Cartier divisor M on $\widetilde{X}$. Since g and $\widetilde{h}$ are isomorphisms over $\widetilde{X}\backslash C'$, we have

$$N\sum_i \lambda_i \widetilde{L}_i|_{\widetilde{h}^{-1}(\widetilde{X}\backslash C')} \sim M|_{\widetilde{X}\backslash C'}.$$

Let $x' \in C'$ be a closed point, let $x := q(x')$, and let $\Omega \subset \widetilde{X}$ be a contractible open neighborhood of x' in the analytic topology. Then $c_1(\mathcal{O}_{\widetilde{X}}(M)|_\Omega) = 0$, so

$$\sum_i \lambda_i c_1\left(\widetilde{\mathcal{L}}_i|_{\Omega\backslash C'}\right) = 0 \in H^2(\Omega\backslash C', \mathbb{Q}).$$

Since q is a local analytic isomorphism at x', by **Step 3**, the image of $c_1\left(\widetilde{\mathcal{L}}_i|_{\Omega\backslash C'}\right)$ in $\mathbb{V}_{C,x}$ is $\tau(c_1(\mathcal{L}_i|_{F_x}))$ for any i. By **Step 2**, $\lambda_i = 0$ for any i.

Step 5. In this step we conclude the proof. Since X' is terminal, X' is smooth near the generic point of T_k for any k. We let E_k' be the prime divisor obtained by the ordinary blow-up of X' at T_k. Then $E_1', \dots, E_m'$ are distinct and $a(E_k', X', 0) = 2$. Since $K_{X'} = g^*K_{\widetilde{X}}$, we have

$$a(E_k', \widetilde{X}, 0) = 2 \quad \text{and} \quad \operatorname{center}_{\widetilde{X}} E_k' = C'$$

for any k. The proposition follows from **Step 4** and Lemma 6.5. □

Set-up 6.8. $X, X', B', h, S_1, \dots, S_m, S, S^\nu, \nu, r_T$ are as follows.

(1) X is a $\mathbb{Q}$-factorial quasi-projective variety of dimension 4.
(2) (X', B') is a $\mathbb{Q}$-factorial terminal pair of dimension 4.
(3) $h : X' \to X$ is a birational morphism.
(4) $S_1, \dots, S_m$ are the h-exceptional prime divisors.
(5) For any k, $h(S_k)$ is a closed point and S_k is an irreducible component of B'.
(6) $S := \bigcup_{k=1}^m S_k$ with the reduced structure.
(7) $\nu : S^\nu := \bigsqcup_{k=1}^m S_k^\nu \to S$ is the normalization of S.
(8) For any irreducible surface $T \subset S$, r_T is the number of prime divisors on S^ν whose image on S is T.

Lemma 6.9. *Notation and conditions as in Set-up 6.8. Then $S = h^{-1}(h(S))$, S is projective, and h is an isomorphism over $X\backslash h(S)$.*

Proof. It follows from [BCHM10, Lemma 3.6.2(3)]. □

Lemma 6.10 (cf. [OAI26b, Lemma 5.2])**.** *Notation and conditions as in Set-up 6.8. Then $r_T = 1$ for all but finitely many irreducible surfaces $T \subset S$, and*

$$\sum_{T\subset S}(r_T - 1) - \sum_{k=1}^m \rho_2(S_k^\nu) \leq \dim \operatorname{Gr}_4^W H^5(S, \mathbb{Q}),$$

where T runs over the irreducible surfaces in S.

Proof. Let $V \subset S$ be a closed subset such that $\dim V \leq 2$ and ν is an isomorphism over $S\backslash V$, and let $F := \nu^{-1}(V)$ with the reduced structure. If an irreducible surface $T \subset S$ is not contained in V, then $r_T = 1$. Since the morphism from the cone of $\mathbb{Q}_S \to \nu_*\mathbb{Q}_{S^\nu}$ to the cone of $\mathbb{Q}_V \to (\nu|_F)_*\mathbb{Q}_F$ induced by the restriction to V is an isomorphism [Sai90, (4.4.1) and (4.4.3)], there exists a distinguished triangle of complexes of mixed Hodge modules

$$\mathbb{Q}_S \to \nu_*\mathbb{Q}_{S^\nu} \oplus \mathbb{Q}_V \to \mathbb{Q}_F \xrightarrow{+1},$$

where the complexes on V and F are pushed forward to S. Taking cohomology and dualizing, we obtain an exact sequence of mixed Hodge structures [Sai89, 1.2 and 1.4]

$$H_5(S, \mathbb{Q}) \xrightarrow{\partial} H_4(F, \mathbb{Q}) \xrightarrow{\alpha} H_4(V, \mathbb{Q}) \oplus H_4(S^\nu, \mathbb{Q}),$$

where the homology groups carry the dual mixed Hodge structures. Since $\dim F \leq 2$ and $\dim V \leq 2$, the classes of the irreducible surfaces in F (resp. V) form a basis of $H_4(F, \mathbb{Q})$ (resp. $H_4(V, \mathbb{Q})$) [Ful98, Lemma 19.1.1], and $H_4(F, \mathbb{Q})$ is pure of weight -4, since the classes of the irreducible surfaces in F are of type $(-2, -2)$ [Sai89, 1.15].

The irreducible surfaces in F are the prime divisors on S^ν whose images on S are contained in V, so

$$\dim H_4(F, \mathbb{Q}) - \dim H_4(V, \mathbb{Q}) = \sum_{T \subset V} (r_T - 1) = \sum_{T \subset S} (r_T - 1).$$

By Lemma 6.9, S^ν_k is projective for any k, so the image of $H_4(F, \mathbb{Q})$ in $H_4(S^\nu, \mathbb{Q})$ is contained in

$$\bigoplus_{k=1}^{m} H_4^{BM,\mathrm{alg}}(S^\nu_k, \mathbb{Q}),$$

whose dimension is $\sum_{k=1}^m \rho_2(S^\nu_k)$ (Definition 2.7). Thus

$$\dim \operatorname{Im}(\partial) = \dim \ker(\alpha) \geq \sum_{T \subset S} (r_T - 1) - \sum_{k=1}^{m} \rho_2(S^\nu_k).$$

Since $H_4(F, \mathbb{Q})$ is pure of weight -4 and ∂ is a morphism of mixed Hodge structures, by [Del71, Théorème 2.3.5], we have

$$\dim \operatorname{Im}(\partial) \leq \dim \operatorname{Gr}^W_{-4} H_5(S, \mathbb{Q}) = \dim \operatorname{Gr}^W_4 H^5(S, \mathbb{Q}).$$

The lemma follows. □

Lemma 6.11 (cf. [OAI26b, Lemma 5.3])**.** *Notation and conditions as in Set-up 6.8. Then* $H^1(S, \mathrm{IC}_{X'}|_S)$ *is pure of weight* 5*.*

Proof. By Lemma 6.9, h is an isomorphism over $X\backslash h(S)$. Thus, by the decomposition theorem [Sai89, Corollary 1.8, Proposition 1.9, and Theorems 1.12 and 2.3],

$$Rh_* \mathrm{IC}_{X'} \cong \mathrm{IC}_X \oplus \bigoplus_{x \in h(S)} \bigoplus_i M_{x,i}[-i],$$

where $M_{x,i}$ is a pure Hodge structure of weight $4 + i$ supported at x. Let $x \in h(S)$. By proper base change [Sai90, (4.4.3)] and Lemma 6.3(1),

$$H^1(h^{-1}(x), \mathrm{IC}_{X'}|_{h^{-1}(x)}) \cong \mathcal{H}^1(Rh_* \mathrm{IC}_{X'})_x \cong \mathcal{H}^1(\mathrm{IC}_X)_x \oplus M_{x,1} = M_{x,1},$$

which is pure of weight 5. By Lemma 6.9, $S = \bigsqcup_{x \in h(S)} h^{-1}(x)$. The lemma follows. □

Lemma 6.12 (cf. [OAI26b, Lemma 5.4])**.** *Notation and conditions as in Set-up 6.8. Then*

$$\dim \operatorname{Gr}^W_4 H^5(S, \mathbb{Q}) \leq \sum_C \beta_C,$$

where C *runs over the irreducible curves in* $(\operatorname{Sing} X') \cap S$*.*

Proof. Let X'_{reg} be the non-singular locus of X'. There exists a distinguished triangle of complexes of mixed Hodge modules

$$\mathbb{Q}_{X'}[4] \to \mathrm{IC}_{X'} \to \mathcal{G} \xrightarrow{+1},$$

where $\mathcal{G}$ is the cone of $\mathbb{Q}_{X'}[4] \to \mathrm{IC}_{X'}$ and the first morphism is the identity on X'_{reg} [Sai89, Proposition 1.14]. Restricting the triangle to S, we obtain an exact sequence of mixed Hodge structures [Sai89, 1.2 and 1.4]

$$H^0(S, \mathcal{G}|_S) \to H^5(S, \mathbb{Q}) \to H^1(S, \mathrm{IC}_{X'}|_S).$$

By Lemma 6.11, $H^1(S, \mathrm{IC}_{X'}|_S)$ has no weight 4 part, so $W_4H^5(S, \mathbb{Q})$ is contained in the image of $H^0(S, \mathcal{G}|_S)$, and so

$$\dim \mathrm{Gr}_4^W H^5(S, \mathbb{Q}) \leq \dim H^0(S, \mathcal{G}|_S).$$

Since $\mathbb{Q}_{X'}[4] \to \mathrm{IC}_{X'}$ is an isomorphism over X'_{reg}, $\mathcal{G}$ is supported on $\operatorname{Sing} X'$. For any $q \geq -2$, we have the exact sequence

$$0 = \mathcal{H}^q(\mathbb{Q}_{X'}[4]) \to \mathcal{H}^q(\mathrm{IC}_{X'}) \to \mathcal{H}^q(\mathcal{G}) \to \mathcal{H}^{q+1}(\mathbb{Q}_{X'}[4]) = 0,$$

hence $\mathcal{H}^q(\mathcal{G}) \cong \mathcal{H}^q(\mathrm{IC}_{X'})$. Since (X', B') is terminal, $\dim \operatorname{Sing} X' \leq 1$. By Lemma 6.3(2), for any irreducible curve $C \subset (\operatorname{Sing} X') \cap S$, with C° and $\mathbb{V}_C$ as in Definition-Lemma 6.6 for X', possibly shrinking C°, we have $\mathcal{H}^q(\mathcal{G})|_{C^\circ} = 0$ for any $q \geq -1$ and $\mathcal{H}^{-2}(\mathcal{G})|_{C^\circ} \cong \mathbb{V}_C$, so

$$H^0_c(C^\circ, \mathcal{G}|_{C^\circ}) \cong H^2_c(C^\circ, \mathbb{V}_C),$$

where H^i_c denotes cohomology with compact supports. Possibly shrinking C°, we may assume $\{C^\circ\}_{C \subset \operatorname{Sing} X' \cap S}$ are disjoint and each C° is open in $\operatorname{Sing} X' \cap S$. Let

$$\Phi := (\operatorname{Sing} X') \cap S \setminus \bigcup_{C \subset \operatorname{Sing} X' \cap S} C^\circ.$$

Then Φ is a finite set of closed points, and

$$H^0(\Phi, \mathcal{G}|_\Phi) = 0$$

by Lemma 6.3(1). Since $\mathcal{G}|_S$ is supported on $(\operatorname{Sing} X') \cap S$, which is projective by Lemma 6.9, we have an exact sequence

$$\bigoplus_C H^2_c(C^\circ, \mathbb{V}_C) \cong \bigoplus_C H^0_c(C^\circ, \mathcal{G}|_{C^\circ}) \to H^0(S, \mathcal{G}|_S) \to H^0(\Phi, \mathcal{G}|_\Phi) = 0.$$

Thus

$$\dim H^0(S, \mathcal{G}|_S) \leq \sum_C \dim H^2_c(C^\circ, \mathbb{V}_C) = \sum_C \beta_C$$

and the lemma follows. □

Theorem 6.13 (cf. [OAI26b, Theorem 5.1])**.** *Notation and conditions as in Set-up 6.8. Then*

$$\sum_{T \subset S} (r_T - 1) \leq \sum_{k=1}^m \rho_2(S_k^\nu) + \#\{E \mid \operatorname{center}_{X'} E \text{ is a curve contained in } S,\ a(E, X', B') < 2\},$$

where T runs over the irreducible surfaces in S and E runs over the prime divisors over X'.

Proof. By Lemmas 6.10 and 6.12,

$$\sum_{T \subset S} (r_T - 1) \leq \sum_{k=1}^m \rho_2(S_k^\nu) + \sum_C \beta_C,$$

where T runs over the irreducible surfaces in S and C runs over the irreducible curves in $(\operatorname{Sing} X') \cap S$. Let C be such a curve. By Proposition 6.7, there exist $\geq \beta_C$ distinct prime divisors E over X' such that $\operatorname{center}_{X'} E = C$ and $a(E, X', 0) = 2$. Since $C \subset S_k$ for some k and S_k is an irreducible component of B', we have

$$a(E, X', B') < a(E, X', 0) = 2$$

for any such E. The theorem follows. □

6.2. **Proof of Proposition 6.1.**

Proof of Proposition 6.1. We let V be the set of integers k such that S_k is exceptional/X. For any i and any irreducible surface $T \subset X'_i$, we define

$$D_{i,T} := \sum_{E|\mathrm{center}_{X'_i} E=T} w_\Gamma(2 - a(E, X'_i, B'_i)) - \sum_{k=1}^{m} r_{i,T,k} d_{i,k} \quad \text{and} \quad r^V_{i,T} := \sum_{k\in V} r_{i,T,k}.$$

Claim 6.14. *For any irreducible surface $T \subset X'_i$, we have*

$$D_{i,T} \geq -\max\{r^V_{i,T} - 1, 0\}. \tag{6.1}$$

Proof. Since (X'_i, B'_i) is terminal, X'_i is terminal, so X'_i is smooth near the generic point of T. Let $m_k := \mathrm{mult}_T S_{i,k}$ for any k.

Assume that $T \not\subset S_{i,k}$ for any k. Then $r_{i,T,k} = 0$ for any k, so $r^V_{i,T} = 0$, so

$$D_{i,T} \geq 0 = -\max\{r^V_{i,T} - 1, 0\}.$$

Thus we may assume that $T \subset S_{i,k}$ for some k.

Let E be the prime divisor obtained by the ordinary blow-up of X'_i at T. Then we have

$$a(E, X'_i, B'_i) = 2 - \sum_{k=1}^{m} m_k b_{i,k}.$$

Since (X'_i, B'_i) is terminal, $a(E, X'_i, B'_i) > 1$, so $\sum_{k=1}^m m_k b_{i,k} < 1$. If S_k is not exceptional/X, then S_k is an irreducible component of B, so $b_{i,k} \in \Gamma$. By Lemma 5.7, there exist distinct prime divisors $E_1, \dots, E_N$ over X'_i such that $\mathrm{center}_{X'_i} E_j = T$ for any j, and

$$\sum_{j=1}^{N} w_\Gamma(2 - a(E_j, X'_i, B'_i)) \geq \sum_{k=1}^{m} r_{i,T,k} d_{i,k} - \max\{r^V_{i,T} - 1, 0\}.$$

The claim follows. □

Proof of Proposition 6.1 continued. By Proposition 5.8(2), $D_j \geq D_{j'}$ for any $j' \geq j \gg 0$. For any $j \gg 0$, there exists i such that $n_i \geq j$, so $D_j \geq D_{n_i}$. Thus we only need to show that $D_{n_i} \geq 0$ for any $i \gg 0$.

Fix i. We let $S := \bigcup_{k\in V} S_{n_i,k}$. Since h_i is a $\mathbb{Q}$-factorial terminalization of (X_i, B_i) and $\{\phi_i\}$ is a sequence of very sloppy flips, the h_i-exceptional prime divisors are exactly $S_{n_i,k}$ for $k \in V$, and for any $k \in V$, we have that $h_i(S_{n_i,k}) = \mathrm{center}_{X_i} S_k$ is a closed point and

$$a(S_k, X_i, B_i) < \lim_{j\to+\infty} a(S_k, X_j, B_j) \leq 1.$$

By Construction-Proposition 4.8(4)(7), $\mathrm{mult}_{S_{n_i,k}} B'_{n_i} > 0$ for any $k \in V$. Now the conditions of Set-up 6.8 are satisfied for $h_i : X'_{n_i} \to X_i$, (X'_{n_i}, B'_{n_i}), and $S_{n_i,k}$ for $k \in V$, and $r_T = r^V_{n_i,T}$ for any irreducible surface $T \subset S$. We let

$$\mathcal{E} := \{E \mid \mathrm{center}_{X'_{n_i}} E \subset S, \dim \mathrm{center}_{X'_{n_i}} E = 1, a(E, X'_{n_i}, B'_{n_i}) < 2\}.$$

By Lemma 5.3(1), $\mathcal{E}$ is a finite set. We let $\kappa := \#\mathcal{E}$. By Theorem 6.13,

$$\sum_{T\subset S} (r^V_{n_i,T} - 1) \leq \sum_{k\in V} \rho_2(S^\nu_{n_i,k}) + \kappa. \tag{6.2}$$

Now we have the following:

(1) For any $E \in \mathcal{E}$, $w_\Gamma(2 - a(E, X'_{n_i}, B'_{n_i})) \geq 1$.

(2) Since $r^V_{n_i,T}=0$ for any irreducible surface $T\not\subset S$ and $r^V_{n_i,T}\geq 1$ for any irreducible surface $T\subset S$, by (6.1),

$$\sum_T\left(\sum_{E|\text{center}_{X'_{n_i}}E=T} w_\Gamma(2-a(E,X'_{n_i},B'_{n_i}))-\sum_{k=1}^m r_{n_i,T,k}d_{n_i,k}\right)\geq -\sum_{T\subset S}(r^V_{n_i,T}-1).$$

(3) For any $k\in V$, since $b_{n_i,k}>0$, we have $d_{n_i,k}\geq 1$ by Definition-Lemma 5.2(1), so

$$\sum_{k=1}^m d_{n_i,k}\rho_2(S^\nu_{n_i,k})\geq\sum_{k\in V}\rho_2(S^\nu_{n_i,k}).$$

By the definition of difficulties (Definition-Proposition 5.5) and (6.2),

$$D_{n_i}\geq\kappa-\sum_{T\subset S}(r^V_{n_i,T}-1)+\sum_{k\in V}\rho_2(S^\nu_{n_i,k})\geq 0.$$

The proposition follows. □

7. Proof of the main theorem

Proof of Theorem 1.3. Since all lc flips terminates in dimension 3 [Sho96], by Proposition 3.2, we may assume that (X,B) is $\mathbb{Q}$-factorial klt. Let $X\to U'\to U$ be the Stein factorization of the associated morphism $\pi: X\to U$. Possibly replacing U with U', we may assume that π is a contraction. Possibly truncating the MMP, we only need to show that any sequence of (K_X+B)-flips/U terminates. By Lemma 4.3, we only need to show that any sequence of sloppy (K_X+B)-flips/U terminates. By Lemmas 4.5 and 4.6, we only need to show that any sequence of very sloppy (K_X+B)-flips/U terminates.

Suppose that there exists an infinite sequence of very sloppy (K_X+B)-flips/U

$$\phi_i:(X_i,B_i)\dashrightarrow(X_{i+1},B_{i+1}),\quad (X_0,B_0):=(X,B).$$

We write

$$\phi_i: X_i\xrightarrow{f_i} Z_i\xleftarrow{f_i^+} X_{i+1}$$

where f_i is the flipping contraction and f_i^+ is the flipped contraction. Now the conditions of Set-up 4.7 are satisfied, and we may apply Construction-Proposition 4.8 and define notation as in Set-up 5.4. In the following we let the notation and conditions be as in Set-up 4.7, Construction-Proposition 4.8, and Set-up 5.4. We let D_i be the difficulty of (X'_i,B'_i) defined in Definition-Proposition 5.5.

By Proposition 3.4, there exists a strictly increasing sequence $\{i_j\}_{j=1}^{+\infty}$ such that $\dim \mathrm{Ex}(f^+_{i_j-1})=2$. We let $V_{i_j}\subset\mathrm{Ex}(f^+_{i_j-1})$ be an irreducible closed subset such that $\dim V_{i_j}=2$ and let η_{i_j} be the generic point of V_{i_j}. By Lemma 4.6, possibly truncating the MMP, we may assume that X_{i_j} is smooth near η_{i_j} for any j. Let E_{i_j} be the prime divisor over X_{i_j} obtained by the ordinary blow-up of X_{i_j} at V_{i_j}. Then

$$a(E_{i_j},X'_{n_{i_j}},B'_{n_{i_j}})=a(E_{i_j},X_{i_j},B_{i_j})=2-t$$

for some $t\in\Gamma$, and $a(E_{i_j},X'_{n_i},B'_{n_i})<2-t$ for any $i<i_j$. By Proposition 5.8(4), possibly truncating the MMP, $D_{n_{i_j-1}}-D_{n_{i_j}}\geq 1$ for any j. By Proposition 5.8(2), possibly truncating the MMP, we have that D_i is decreasing and $\lim_{i\to+\infty}D_i=-\infty$. This contradicts Proposition 6.1. □

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